EN.553.632 Bayesian Statistics

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Chapter 1

Introduction and Probability Review

1.1 Bayesian vs Frequentist Perspectives

Definition 1.1.1. Probability can be interpreted in two fundamentally different ways:

- **Frequentist:** Probability is the long-run relative frequency of an event.
- **Bayesian:** Probability is a degree of belief about an uncertain quantity.

In the Bayesian framework, unknown parameters are treated as random variables. This contrasts with the frequentist view, where parameters are fixed but unknown.

Example 1.1.1. Consider a coin flipped $n = 10$ times with 7 heads observed.

- Frequentist estimate: $\hat{\theta} = 0.7$
- Bayesian approach: place a prior $\theta \sim \text{Beta}(\alpha, \beta)$

After observing $y = 7$ heads, the posterior becomes:

$$\theta \mid y \sim \text{Beta}(\alpha + y, \beta + n - y)$$

1.2 Bayes' Theorem

Theorem 1.2.1 (Bayes' Rule)**.** *For parameter θ and data y,*

$$p(\theta \mid y) = \frac{p(y \mid \theta)p(\theta)}{p(y)}$$

where

$$p(y) = \int p(y \mid \theta)p(\theta)\,\mathrm{d}\theta$$

Remark 1.2.1. The posterior is often written up to proportionality:

$$p(\theta \mid y) \propto p(y \mid \theta)p(\theta)$$

1.3 Bernoulli and Binomial Models

1.3.1 Bernoulli Model

Definition 1.3.1. A Bernoulli random variable satisfies:

$$Y \sim \text{Bernoulli}(\theta), \quad P(Y = 1) = \theta$$

1.3.2 Binomial Model

Definition 1.3.2. For n independent Bernoulli trials,

$$Y \sim \text{Binomial}(n, \theta), \quad P(Y = y|\theta) = \binom{n}{y}\theta^y(1-\theta)^{n-y}$$

1.3.3 Likelihood

Given observations $y_1, \ldots, y_n$, the likelihood is:

$$L(\theta) = \prod_{i=1}^{n} \theta^{y_i}(1-\theta)^{1-y_i} = \theta^{\sum y_i}(1-\theta)^{n-\sum y_i}$$

1.3.4 Maximum Likelihood Estimation

Proposition 1.3.1. *The MLE for θ in the Bernoulli/Binomial model is:*

$$\hat{\theta}_{MLE} = \bar{y}$$

Proof. The log-likelihood is:

$$\ell(\theta) = \sum y_i \log\theta + (n - \sum y_i)\log(1-\theta)$$

Differentiate:

$$\frac{d\ell}{d\theta} = \frac{\sum y_i}{\theta} - \frac{n - \sum y_i}{1-\theta}$$

Set equal to zero:

$$\frac{\sum y_i}{\theta} = \frac{n - \sum y_i}{1-\theta} \Rightarrow \theta = \frac{1}{n}\sum y_i$$

□

1.4 Conjugate Priors: Beta-Binomial Model

Definition 1.4.1. A prior is conjugate if the posterior belongs to the same family as the prior.

Proposition 1.4.1. *If*

$$\theta \sim \text{Beta}(\alpha, \beta), \quad Y \mid \theta \sim \text{Binomial}(n, \theta)$$

then

$$\theta \mid y \sim \text{Beta}(\alpha + y, \beta + n - y)$$

Proof. Posterior:

$$p(\theta \mid y) \propto \theta^y (1-\theta)^{n-y} \cdot \theta^{\alpha-1}(1-\theta)^{\beta-1}$$

Combine exponents:

$$p(\theta \mid y) \propto \theta^{\alpha+y-1}(1-\theta)^{\beta+n-y-1}$$

which is the kernel of a Beta distribution. □

1.4.1 Posterior Mean

$$\mathbb{E}[\theta \mid y] = \frac{\alpha + y}{\alpha + \beta + n}$$

Remark 1.4.1. This can be interpreted as a weighted average:

$$\mathbb{E}[\theta \mid y] = w \cdot \frac{\alpha}{\alpha + \beta} + (1 - w) \cdot \frac{y}{n}$$

1.5 Credible Intervals and HPD Regions

Definition 1.5.1. An interval (a, b) is a $(1 - \alpha)$ credible interval if

$$\mathbb{P}(a < \theta < b \mid y) = 1 - \alpha$$

Definition 1.5.2. A highest posterior density (HPD) region A satisfies:

- $\mathbb{P}(\theta \in A \mid y) = 1 - \alpha$
- $p(\theta_1 \mid y) > p(\theta_2 \mid y)$ for $\theta_1 \in A$, $\theta_2 \notin A$

1.6 Posterior Predictive Distribution

Proposition 1.6.1. *For a new observation Z,*

$$p(Z = 1 \mid y) = \mathbb{E}[\theta \mid y]$$

Proof.

$$p(Z = 1 \mid y) = \int p(Z = 1 \mid \theta) p(\theta \mid y) \, \mathrm{d}\theta = \int \theta p(\theta \mid y) \, \mathrm{d}\theta = \mathbb{E}[\theta \mid y]$$

□

1.7 Exponential Family

Definition 1.7.1. A distribution belongs to the exponential family if it can be written as:

$$p(y \mid \theta) = f(y) g(\theta) \exp\{\phi(\theta) u(y)\}$$

Remark 1.7.1. Many common distributions (Normal, Binomial, Poisson) belong to this family, which enables conjugate prior constructions.

1.8 Prior Specification

1.8.1 Subjective Priors

Subjective priors encode information available before the current data. A useful elicitation asks experts about observable or interpretable quantities, such as a plausible median, central range, or prior predictive event count, and then solves for hyperparameters. Eliciting a variance parameter directly is often less reliable than eliciting the range of future observations it implies.

For example, a beta prior can be chosen by specifying a prior mean m and an effective sample size s: set $a = ms$ and $b = (1 - m)s$. The choice separates location from concentration and can be checked through the beta–binomial prior predictive distribution.

1.8.2 Non-informative Priors

There is no prior that is uniformly noninformative under every parameterization. A density flat in θ is not flat in $\phi = g(\theta)$ because densities transform by a Jacobian. Improper priors such as $\pi(\theta) \propto 1$ may be useful reference limits, but posterior propriety must be proved and ordinary Bayes factors cannot use their arbitrary normalizing constants.

1.8.3 Jeffreys Prior

Jeffreys' rule uses Fisher information,

$$\pi_J(\theta) \propto \sqrt{I(\theta)}, \qquad I(\theta) = -\mathbb{E}_\theta\left[\frac{\partial^2}{\partial\theta^2}\log p(Y \mid \theta)\right].$$

If $\phi = g(\theta)$ is one-to-one, Fisher information transforms so that $\pi_J(\phi) = \pi_J(\theta)|\,\mathrm{d}\theta/\,\mathrm{d}\phi|$; hence the induced prior is invariant to smooth reparameterization.

Example 1.8.1 (Bernoulli Jeffreys prior)**.** For one Bernoulli observation, $I(\theta) = 1/\{\theta(1-\theta)\}$. Therefore

$$\pi_J(\theta) \propto \theta^{-1/2}(1-\theta)^{-1/2},$$

the Beta$(1/2, 1/2)$ density. It is proper but places more mass near the boundaries than the uniform prior. In multiparameter or hierarchical models, blind application of Jeffreys' rule can produce an improper posterior, so invariance does not remove the need for model-specific checking.

1.9 Poisson-Gamma Model

Let $Y_1, \ldots, Y_n \mid \theta \overset{\text{iid}}{\sim} \text{Poisson}(\theta)$, where θ is the expected count per observational unit. The likelihood is

$$L(\theta; y) = \prod_{i=1}^{n} \frac{\theta^{y_i} e^{-\theta}}{y_i!} \propto \theta^S e^{-n\theta}, \qquad S = \sum_{i=1}^{n} y_i.$$

Thus S is sufficient for θ. We use the shape–rate convention

$$\theta \sim \text{Gamma}(a, b), qquad p(\theta) = \frac{b^a}{\Gamma(a)}\theta^{a-1}e^{-b\theta},$$

so that $\mathbb{E}(\theta) = a/b$ and $\text{Var}(\theta) = a/b^2$. Stating the convention is essential because some texts use a scale parameter instead of a rate.

Proposition 1.9.1 (Gamma–Poisson conjugacy)**.** *The posterior is*

$$\theta \mid y \sim \text{Gamma}(a + S, b + n).$$

Proof. Multiplying likelihood and prior kernels gives

$$p(\theta \mid y) \propto \theta^S e^{-n\theta}\theta^{a-1}e^{-b\theta} = \theta^{a+S-1}e^{-(b+n)\theta},$$

which is the kernel of the stated gamma distribution. □

The posterior mean has the precision-like weighted form

$$\mathbb{E}(\theta \mid y) = \frac{b}{b+n}\frac{a}{b} + \frac{n}{b+n}\bar{y}.$$

Here b behaves like a prior exposure and a like a prior event count. This interpretation is exact algebraically, although a and b need not be integers.

For a future count $\widetilde{Y} \mid \theta \sim \text{Poisson}(\theta)$, integrating over the posterior gives a negative-binomial predictive law,

$$\mathbb{P}(\widetilde{Y} = k \mid y) = \frac{\Gamma(a+S+k)}{\Gamma(a+S)k!} \left(\frac{b+n}{b+n+1}\right)^{a+S} \left(\frac{1}{b+n+1}\right)^{k}.$$

Its variance exceeds its mean because it includes both future Poisson variation and posterior uncertainty in θ.

Example 1.9.1 (Event rate updating)**.** Suppose $n = 10$ equal-exposure units produce $S = 24$ events and the prior is Gamma$(2, 1)$. Then $\theta \mid y \sim \text{Gamma}(26, 11)$, with posterior mean $26/11 \approx 2.36$. The sample mean is 2.4 and the prior mean is 2, so the posterior mean lies between them. A plug-in Poisson predictive distribution would have variance 2.36, whereas the posterior predictive variance is $2.36 + 26/11^2 \approx 2.58$.

1.9.1 Unequal exposures

If unit i is observed for exposure e_i, use $Y_i \mid \theta \sim \text{Poisson}(e_i\theta)$. The likelihood kernel becomes

$$L(\theta) \propto \theta^{\sum_i y_i} \exp\left(-\theta \sum_i e_i\right),$$

so a Gamma(a, b) prior updates to

$$\theta \mid y \sim \text{Gamma}\left(a + \sum_i y_i, b + \sum_i e_i\right).$$

Counts contribute to the shape and time at risk contributes to the rate. The posterior mean is therefore a compromise between the prior event rate and the observed total count divided by total exposure, not the unweighted average of the individual counts.

Example 1.9.2. Two sites report 4 events over 10 person-years and 6 events over 30 person-years. Their raw counts suggest that the second site has more events, but the pooled exposure-adjusted rate is $10/40 = 0.25$. With prior Gamma$(1, 2)$, the posterior is Gamma$(11, 42)$ and has mean 0.262 events per person-year.

1.10 Preview: Computational Methods

Conjugacy is exceptional. In later chapters, posterior expectations are approximated by independent Monte Carlo when direct draws are available, importance sampling when a tractable proposal covers the target, and MCMC when only an unnormalized posterior kernel can be evaluated. These methods solve different computational problems and have different error diagnostics.

1.11 Common Distributions and Identities

1.11.1 Beta and Gamma Functions

Definition 1.11.1. The Gamma function is defined as:

$$\Gamma(a) = \int_0^\infty x^{a-1} e^{-x} \, \mathrm{d}x$$

and satisfies:

$$\Gamma(n) = (n-1)! \quad \text{for integer } n$$

Definition 1.11.2. The Beta distribution has density:

$$f(\theta) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \theta^{a-1}(1-\theta)^{b-1}$$

Proposition 1.11.1. *For $\theta \sim \text{Beta}(a, b)$:*

$$\mathbb{E}[\theta] = \frac{a}{a+b}, \quad \text{Var}(\theta) = \frac{ab}{(a+b)^2(a+b+1)}$$

1.11.2 Gamma Distribution

Definition 1.11.3. A Gamma distribution with shape a and rate b:

$$f(\theta) = \frac{b^a}{\Gamma(a)} \theta^{a-1} e^{-b\theta}$$

Proposition 1.11.2.

$$\mathbb{E}[\theta] = \frac{a}{b}, \quad \text{Var}(\theta) = \frac{a}{b^2}$$

Remark 1.11.1. The Gamma distribution is conjugate to the Poisson likelihood.

1.12 Sufficiency and Likelihood Structure

Definition 1.12.1. A statistic $T(y)$ is sufficient for θ if the likelihood can be written as:

$$p(y \mid \theta) = g(T(y), \theta) h(y)$$

Proposition 1.12.1. *For:*

- *Binomial model: sufficient statistic is $y = \sum y_i$*
- *Poisson model: sufficient statistic is $\sum y_i$*

Remark 1.12.1. This reduction is critical in Bayesian updating since posterior distributions depend only on sufficient statistics.

1.13 Posterior as a Weighted Average

A key Bayesian insight is that posterior estimates combine prior and data:

Proposition 1.13.1. *For the Beta-Binomial model:*

$$\mathbb{E}[\theta \mid y] = \frac{a+y}{a+b+n}$$

can be written as:

$$\mathbb{E}[\theta \mid y] = \frac{a+b}{a+b+n} \cdot \frac{a}{a+b} + \frac{n}{a+b+n} \cdot \frac{y}{n}$$

Remark 1.13.1. This shows:

- Prior contributes as pseudo-observations
- Data influence increases with n

1.14 Bias, Variance, and MSE

Definition 1.14.1. Bias of an estimator $\hat{\theta}$:

$$\text{Bias} = \mathbb{E}[\hat{\theta}] - \theta$$

Definition 1.14.2. Mean Squared Error:

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + \text{Bias}^2$$

Remark 1.14.1. Bayesian estimators are often biased but achieve lower MSE through shrinkage.

1.15 Posterior Predictive Distribution

Definition 1.15.1. The posterior predictive distribution is:

$$p(\tilde{y} \mid y) = \int p(\tilde{y} \mid \theta) p(\theta \mid y) \, \mathrm{d}\theta$$

Remark 1.15.1. This integrates out parameter uncertainty, a key advantage over plug-in estimators.

1.16 Exponential Family and Conjugacy

In canonical form, a k-parameter exponential family has density

$$p(y \mid \eta) = h(y) \exp\{\eta^\top T(y) - A(\eta)\},$$

where η is the natural parameter and A is the log-partition function. For an IID sample, the likelihood depends on the data only through $\sum_i T(Y_i)$, proving sufficiency by the factorization theorem.

Differentiating the identity $\int h(y) e^{\eta^\top T(y) - A(\eta)} \, \mathrm{d}y = 1$ gives

$$\nabla A(\eta) = \mathbb{E}_\eta\{T(Y)\}, \qquad \nabla^2 A(\eta) = \mathrm{Cov}_\eta\{T(Y)\}.$$

Thus A is convex, and its curvature is the Fisher information in natural coordinates.

The generic conjugate prior is

$$p(\eta \mid \xi, \kappa) \propto \exp\{\eta^\top \xi - \kappa A(\eta)\}.$$

After observing $y_1, \ldots, y_n$, its hyperparameters update as

$$\xi_n = \xi + \sum_{i=1}^{n} T(y_i), \qquad \kappa_n = \kappa + n.$$

The familiar beta–binomial, gamma–Poisson, and normal conjugate analyses are instances of this addition of prior and sample sufficient statistics.

1.17 Interpretation of Bayesian Updating

Bayesian updating converts a prior distribution into a posterior distribution; an action or point estimate still requires a loss function. For an action a and loss $L(\theta, a)$, the Bayes action minimizes posterior expected loss,

$$a^*(y) = \arg\min_a \mathbb{E}\{L(\theta, a) \mid y\}.$$

Under squared-error loss the minimizer is the posterior mean, under absolute loss it is a posterior median, and under zero–one loss on a discrete parameter it is a posterior mode. These are different summaries because a posterior distribution does not determine what consequences matter.

Squared-error Bayes action. Writing $m = \mathbb{E}(\theta \mid y)$,

$$\mathbb{E}\{(\theta - a)^2 \mid y\} = \mathrm{Var}(\theta \mid y) + (m - a)^2.$$

The posterior variance is independent of a, so the unique minimizer is $a = m$. □

Example 1.17.1. In rare-event estimation, even if $y = 0$, the posterior remains well-defined:

$$\theta \mid y \sim \mathrm{Beta}(a, b + n)$$

which avoids degenerate estimates like $\hat{\theta} = 0$.

1.18 Prior Predictive Distributions and Calibration

Before observing data, the model implies

$$p(y) = \int p(y \mid \theta) p(\theta) \, \mathrm{d}\theta.$$

Simulating $\theta^{(s)} \sim p(\theta)$ and then $Y^{(s)} \sim p(\cdot \mid \theta^{(s)})$ reveals the observable implications of the prior. A prior that looks diffuse on a parameter scale may be highly informative on the data scale.

Example 1.18.1 (Beta–binomial prior prediction)**.** Suppose

$$\theta \sim \mathrm{Beta}(a, b), \qquad Y \mid \theta \sim \mathrm{Binomial}(n, \theta).$$

Then

$$\mathbb{P}(Y = y) = \binom{n}{y} \frac{B(a + y, b + n - y)}{B(a, b)}.$$

Its mean is $na/(a+b)$ and its variance exceeds that of a binomial with fixed success probability $a/(a+b)$ because uncertainty in θ induces positive dependence among trials. Plotting this distribution is a direct way to decide whether the prior allows plausible success counts.

Prior predictive checking assesses the prior and likelihood before data are used twice. Posterior predictive checking instead asks whether the fitted model reproduces selected features after learning from the observations.

Chapter 2

Bayesian Decision Theory and Large-Sample Behavior

2.1 Decision Rules, Loss, and Risk

A statistical procedure maps data to an action. Let $\delta(Y)$ denote a decision rule and let $L(\theta, a)$ quantify the consequence of action a when the parameter is θ. Its frequentist risk is

$$R(\theta, \delta) = \mathbb{E}_\theta L\{\theta, \delta(Y)\}.$$

Given a prior π, the Bayes risk averages this function over parameter values,

$$r(\pi, \delta) = \int R(\theta, \delta)\pi(\theta)\,\mathrm{d}\theta.$$

Fubini's theorem rewrites the Bayes risk as an expectation over the marginal law of Y of posterior expected loss. Therefore a rule that minimizes $\mathbb{E}\{L(\theta, a) \mid Y = y\}$ separately for every y is a Bayes rule.

Proposition 2.1.1. *For a scalar parameter, the posterior mean minimizes squared-error loss, any posterior median minimizes absolute-error loss, and a posterior α quantile minimizes the asymmetric check loss*

$$L_\alpha(\theta, a) = (\theta - a)\{\alpha - \mathbb{1}(\theta < a)\}.$$

Proof. The squared-error result follows from the posterior bias–variance decomposition. For absolute loss, a subgradient in a is $2F(a \mid y) - 1$, so zero belongs to the subgradient precisely at a posterior median. For check loss, the corresponding subgradient is $F(a \mid y) - \alpha$, which vanishes at an α quantile. □

Example 2.1.1 (Prediction with asymmetric costs)**.** Suppose a posterior predictive demand distribution has 0.9 quantile equal to 120 units. If understocking costs nine times as much per unit as overstocking, the relevant check-loss level is 0.9, and the Bayes action is to stock 120 units. The predictive mean would solve a different decision problem and need not be optimal here.

2.2 Admissibility and Complete Classes

A rule δ_1 dominates δ_0 if its risk is no larger for every θ and strictly smaller for at least one value. A rule is admissible if no other rule dominates it. Under standard regularity conditions, a unique Bayes rule under a proper prior with full support is admissible: if another rule improved its risk on a set with positive prior mass, it would have strictly smaller Bayes risk, contradicting optimality.

The converse is subtler. Complete-class theorems state, under suitable conditions, that every admissible rule is Bayes or a limit of Bayes rules. This gives Bayesian procedures a frequentist decision-theoretic role even when the prior is used as a device for constructing estimators rather than as a literal population distribution.

2.3 Credible Sets and Coverage

A $(1-\alpha)$ posterior credible set $C(Y)$ satisfies

$$\mathbb{P}\{\theta \in C(Y) \mid Y\} = 1 - \alpha.$$

This is a conditional probability statement about the parameter under the posterior. Frequentist coverage instead fixes θ_0 and evaluates

$$\mathbb{P}_{\theta_0}\{\theta_0 \in C(Y)\}$$

over repeated datasets. The two probabilities condition on different quantities and need not agree in finite samples.

Example 2.3.1 (Normal mean)**.** If $Y_1, \dots, Y_n \overset{\text{iid}}{\sim} \mathcal{N}(\theta, \sigma^2)$ with known σ and the flat prior $\pi(\theta) \propto 1$, then

$$\theta \mid y \sim \mathcal{N}(\bar{y}, \sigma^2/n).$$

The equal-tailed posterior interval $\bar{y} \pm z_{1-\alpha/2}\sigma/\sqrt{n}$ is also the classical confidence interval and has exact frequentist coverage. This agreement is a special location-model calculation, not a general identity between credible and confidence intervals.

For a skewed posterior, an equal-tailed interval and a highest-posterior- density region can differ substantially. An HPD region has the form $\{\theta : p(\theta \mid y) \geq c\}$ with c chosen to attain the desired probability. Among sets defined relative to the same dominating measure, it has minimum volume, but HPD regions are not invariant under nonlinear reparameterization because density height changes with the Jacobian.

2.4 Improper Priors and Posterior Propriety

An improper prior does not integrate to one. It may still yield a proper posterior if

$$0 < \int p(y \mid \theta)\pi(\theta)\,\mathrm{d}\theta < \infty.$$

This condition must be checked for the realized design and sample size; Bayes' formula does not repair an arbitrary improper prior automatically.

Example 2.4.1 (Normal location and scale)**.** For $Y_i \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$, the reference prior $\pi(\mu, \sigma^2) \propto 1/\sigma^2$ gives

$$p(\mu, \sigma^2 \mid y) \propto (\sigma^2)^{-(n/2+1)} \exp\left\{-\frac{\sum_i (y_i - \mu)^2}{2\sigma^2}\right\}.$$

After integrating over μ, the remaining integral is finite when $n \geq 2$ and the observations are not all identical. With one observation, location and scale cannot both be learned and the posterior is improper.

Improper priors are especially dangerous in hierarchical models, logistic separation, and variance-component problems. They also cannot be used directly for ordinary marginal likelihoods because their arbitrary multiplicative constants do not cancel between models.

2.5 Posterior Consistency

Let P_{θ_0} generate the data. A posterior is consistent at θ_0 if for every neighborhood U of θ_0,

$$\Pi(U^c \mid Y_1, \ldots, Y_n) \to 0 \quad \text{in } P_{\theta_0}\text{-probability.}$$

In finite-dimensional identifiable IID models, consistency typically follows when the prior assigns positive mass to every Kullback–Leibler neighborhood of the truth and uniformly consistent tests separate the truth from its complement. The prior need not be centered at the truth, but it must not exclude it.

Example 2.5.1 (Beta–Bernoulli consistency)**.** With a Beta(a, b) prior and true success probability $\theta_0 \in (0, 1)$, the posterior is Beta$(a + S_n, b + n - S_n)$. Its mean converges almost surely to θ_0 because $S_n/n \to \theta_0$, while its variance is $O(1/n)$. Chebyshev's inequality then shows that posterior mass outside every fixed neighborhood of θ_0 vanishes.

2.6 Bernstein–von Mises Theorem

In a regular p-dimensional model with true interior parameter θ_0, nonsingular Fisher information $I(\theta_0)$, and a prior density positive and continuous near θ_0, the posterior is asymptotically normal:

$$\mathcal{L}\Big(\sqrt{n}(\theta - \hat{\theta}_n) \mid Y_{1:n}\Big) \longrightarrow \mathcal{N}_p\{0, I(\theta_0)^{-1}\}$$

in total variation under suitable regularity conditions. Here $\hat{\theta}_n$ may be the MLE or another efficient estimator.

The theorem explains why regular Bayesian credible sets often have asymptotically correct frequentist coverage and why the prior contributes only lower-order effects. It does not apply automatically to boundary parameters, nonidentifiable mixtures, increasing-dimensional models, irregular likelihoods, or many nonparametric problems. In those settings the posterior can converge at a nonstandard rate or to a non-Gaussian limit.

Chapter 3

The Binomial Model

3.1 Model Setup

Definition 3.1.1. Let $Y \mid \theta \sim \text{Binomial}(n, \theta)$ with probability mass function:

$$p(y \mid \theta) = \binom{n}{y} \theta^y (1-\theta)^{n-y}, \quad \theta \in [0, 1]$$

Remark 3.1.1. If $Y_i \overset{\text{iid}}{\sim} \text{Bernoulli}(\theta)$, then $Y = \sum_{i=1}^{n} Y_i \sim \text{Binomial}(n, \theta)$. The sufficient statistic is $y = \sum Y_i$.

3.2 Likelihood and Maximum Likelihood Estimation

Ignoring constants:

$$L(\theta) \propto \theta^y (1-\theta)^{n-y}$$

Proposition 3.2.1. *The log-likelihood is:*

$$\ell(\theta) = y \log \theta + (n-y) \log(1-\theta)$$

Proposition 3.2.2. *The MLE for θ is:*

$$\hat{\theta} = \frac{y}{n}$$

Proof. Differentiate:

$$\frac{d\ell}{d\theta} = \frac{y}{\theta} - \frac{n-y}{1-\theta}$$

Set equal to zero:

$$\frac{y}{\theta} = \frac{n-y}{1-\theta} \Rightarrow \theta = \frac{y}{n}$$

□

3.3 Beta Prior

Definition 3.3.1. Assume prior:

$$\theta \sim \text{Beta}(a, b), \quad \pi(\theta) \propto \theta^{a-1}(1-\theta)^{b-1}$$

Remark 3.3.1. The parameters (a, b) can be interpreted as prior pseudo-counts:

$$a - 1 \text{ successes}, \quad b - 1 \text{ failures}$$

3.4 Posterior Distribution

Theorem 3.4.1. *If* $\theta \sim \text{Beta}(a, b)$ *and* $Y \mid \theta \sim \text{Binomial}(n, \theta)$*, then:*

$$\theta \mid y \sim \text{Beta}(a + y, b + n - y)$$

Proof.

$$p(\theta \mid y) \propto \theta^{y}(1-\theta)^{n-y} \cdot \theta^{a-1}(1-\theta)^{b-1}$$

$$= \theta^{a+y-1}(1-\theta)^{b+n-y-1}$$

□

3.5 Posterior Summaries

For $\theta \mid y \sim \text{Beta}(a + y, b + n - y)$, the mean, variance, and interior mode are

$$\mathbb{E}(\theta \mid y) = \frac{a+y}{a+b+n},$$

$$\text{Var}(\theta \mid y) = \frac{(a+y)(b+n-y)}{(a+b+n)^2(a+b+n+1)},$$

and, when $a + y > 1$ and $b + n - y > 1$,

$$\text{mode}(\theta \mid y) = \frac{a+y-1}{a+b+n-2}.$$

The boundary cases matter: if a beta shape parameter is at most one, the density may be monotone or unbounded at zero or one, so the interior formula is not valid.

The posterior mean decomposes as

$$\mathbb{E}[\theta \mid y] = \frac{a+b}{a+b+n} \cdot \frac{a}{a+b} + \frac{n}{a+b+n} \cdot \frac{y}{n}$$

and hence shrinks the sample proportion toward the prior mean. The effective prior sample size is $a + b$ under this mean-based interpretation.

3.6 Posterior Predictive Distribution

Definition 3.6.1. For a new observation $\tilde{Y}$:

$$p(\tilde{Y} = 1 \mid y) = \int \theta p(\theta \mid y)\, \mathrm{d}\theta$$

Proposition 3.6.1.

$$p(\tilde{Y} = 1 \mid y) = \mathbb{E}[\theta \mid y] = \frac{a + y}{a + b + n}$$

3.7 Credible Intervals

An equal-tailed $(1 - \alpha)$ interval uses beta posterior quantiles

$$\left[F^{-1}_{\text{Beta}}(\alpha/2), F^{-1}_{\text{Beta}}(1 - \alpha/2)\right].$$

It places probability $\alpha/2$ in each tail. This is a direct posterior statement conditional on the observed data; its repeated-sampling coverage is a separate property.

3.8 Highest Posterior Density Regions

An HPD set has the form

$$A_c = \{\theta : p(\theta \mid y) \geq c\}, \qquad \mathbb{P}(\theta \in A_c \mid y) = 1 - \alpha.$$

For a unimodal beta posterior it is an interval whose endpoints have equal posterior density. It is shorter than the equal-tailed interval for a skewed posterior, but density height depends on parameterization, so HPD sets are not invariant under nonlinear transformations.

3.9 Shrinkage Interpretation

Writing $w = (a + b)/(a + b + n)$, the posterior mean is $w\, a/(a + b) + (1 - w)y/n$. Small samples are pulled more strongly toward the prior mean; as n grows, $w \to 0$ and the data dominate. Shrinkage concerns a point summary, while the full beta posterior also records remaining uncertainty.

3.10 Rare Event Example

Example 3.10.1. Suppose:

$$n = 20, \quad y = 0, \quad \theta \sim \text{Beta}(2, 20)$$

Then:

$$\theta \mid y \sim \text{Beta}(2, 40)$$

Remark 3.10.1. The MLE gives $\hat{\theta} = 0$, while the Bayesian estimator remains strictly positive.

3.11 Sequential Updating and Beta–Binomial Prediction

Conjugate updating is sequentially coherent. If the first batch contains y_1 successes in n_1 trials, the posterior is $\text{Beta}(a + y_1, b + n_1 - y_1)$. Using this posterior as the prior for a second batch produces

$$\theta \mid y_1, y_2 \sim \text{Beta}(a + y_1 + y_2, b + n_1 + n_2 - y_1 - y_2),$$

the same answer obtained by analyzing both batches together.

For m future trials, integrating the binomial sampling law gives

$$\mathbb{P}(\widetilde{Y} = k \mid y) = \binom{m}{k} \frac{B(a + y + k, b + n - y + m - k)}{B(a + y, b + n - y)}.$$

This beta–binomial law is overdispersed relative to a binomial evaluated at the posterior mean, because future Bernoulli outcomes share the same uncertain θ and are positively correlated after it is integrated out.

Example 3.11.1 (Posterior probability of a clinically relevant rate)**.** With a $\text{Beta}(1, 1)$ prior and 18 responses among 30 patients, the posterior is $\text{Beta}(19, 13)$. Evidence that the response rate exceeds a clinically relevant threshold 0.5 is reported directly as $1 - F_{\text{Beta}(19,13)}(0.5)$. For planning the next ten patients, the beta–binomial predictive distribution is appropriate; a binomial with $p = 19/32$ would omit posterior uncertainty.

3.12 Frequentist vs Bayesian Interpretation

Remark 3.12.1. Frequentist interval:

$$\mathbb{P}(l(Y) < \theta < u(Y) \mid \theta) = 1 - \alpha$$

Bayesian interval:

$$\mathbb{P}(l < \theta < u \mid y) = 1 - \alpha$$

Chapter 4

Poisson Model and Exponential Families

4.1 Poisson Model

Definition 4.1.1. A random variable Y follows a Poisson distribution with rate $\theta > 0$ if:

$$Y \mid \theta \sim \text{Poisson}(\theta), \quad p(y \mid \theta) = \frac{\theta^y e^{-\theta}}{y!}, \quad y \in \mathbb{N}$$

Proposition 4.1.1. *For $Y \sim \text{Poisson}(\theta)$:*

$$\mathbb{E}[Y] = \theta, \quad \text{Var}(Y) = \theta$$

Remark 4.1.1. The equality of mean and variance is a defining feature of the Poisson model. Large deviations between sample mean and variance indicate model misspecification.

4.2 Likelihood and Sufficiency

Let $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \text{Poisson}(\theta)$. The joint likelihood is:

$$p(y_1, \ldots, y_n \mid \theta) = \prod_{i=1}^{n} \frac{\theta^{y_i} e^{-\theta}}{y_i!} = c(y)\, \theta^{\sum y_i} e^{-n\theta}$$

Proposition 4.2.1. *The sufficient statistic is:*

$$T(y) = \sum_{i=1}^{n} y_i$$

Remark 4.2.1. All information about θ is contained in the total count $\sum y_i$.

4.3 Maximum Likelihood Estimation

Proposition 4.3.1. *The MLE for θ is:*

$$\hat{\theta} = \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$$

Proof. Log-likelihood:

$$\ell(\theta) = \sum y_i \log\theta - n\theta$$

Differentiate:

$$\frac{d\ell}{d\theta} = \frac{\sum y_i}{\theta} - n$$

Set to zero:

$$\theta = \frac{1}{n}\sum y_i$$

□

4.4 Gamma Prior

Definition 4.4.1. Assume prior:

$$\theta \sim \text{Gamma}(a,b), \quad \pi(\theta) = \frac{b^a}{\Gamma(a)}\theta^{a-1}e^{-b\theta}$$

Proposition 4.4.1.

$$\mathbb{E}[\theta] = \frac{a}{b}, \quad \text{Var}(\theta) = \frac{a}{b^2}$$

4.5 Posterior Distribution

Theorem 4.5.1. *If $\theta \sim \text{Gamma}(a,b)$ and $Y_i \overset{iid}{\sim} \text{Poisson}(\theta)$, then:*

$$\theta \mid y \sim \text{Gamma}\left(a + \sum y_i,\ b + n\right)$$

Proof.

$$p(\theta \mid y) \propto \theta^{\sum y_i} e^{-n\theta} \cdot \theta^{a-1} e^{-b\theta}$$

$$= \theta^{a+\sum y_i - 1} e^{-(b+n)\theta}$$

□

4.6 Posterior Summaries

4.6.1 Mean

$$\mathbb{E}[\theta \mid y] = \frac{a + \sum y_i}{b + n}$$

Remark 4.6.1. This can be written as:

$$\mathbb{E}[\theta \mid y] = \frac{b}{b+n} \cdot \frac{a}{b} + \frac{n}{b+n} \cdot \frac{\sum y_i}{n}$$

4.6.2 Variance

$$\mathrm{Var}(\theta \mid y) = \frac{a + \sum y_i}{(b+n)^2}$$

4.7 Posterior Predictive Distribution

Definition 4.7.1. For a new observation $\tilde{Y}$:

$$p(\tilde{y} \mid y) = \int p(\tilde{y} \mid \theta) p(\theta \mid y) \, \mathrm{d}\theta$$

Remark 4.7.1. This results in a negative binomial distribution, reflecting overdispersion relative to the Poisson model.

4.8 Example: Birth Rate Comparison

Example 4.8.1. Suppose:

- Group 1: $n_1 = 111$, $\sum y_i = 217$
- Group 2: $n_2 = 44$, $\sum y_i = 66$
- Prior: $\theta \sim \mathrm{Gamma}(2, 1)$

Posteriors:

$$\theta_1 \mid y \sim \mathrm{Gamma}(219, 112), \quad \theta_2 \mid y \sim \mathrm{Gamma}(68, 45)$$

4.9 Exponential Family Representation

Definition 4.9.1. A distribution belongs to the exponential family if:

$$p(y \mid \phi) = h(y) c(\phi) \exp\{\phi t(y)\}$$

4.9.1 Poisson as an Exponential Family

Proposition 4.9.1. *The Poisson distribution can be written as:*

$$p(y \mid \theta) = \frac{1}{y!} \exp\{y \log \theta - \theta\}$$

Remark 4.9.1. This corresponds to:

$$t(y) = y, \quad \phi = \log \theta$$

4.10 General Exponential Family Structure

Definition 4.10.1. A one-parameter exponential family has the form:

$$p(y \mid \phi) = h(y)c(\phi) \exp\{\phi t(y)\}$$

Proposition 4.10.1. *For exponential families:*

- $t(y)$ *is a sufficient statistic*
- *Conjugate priors exist due to exponential structure*

Remark 4.10.1. Common examples:

- Binomial $\leftrightarrow$ Beta
- Poisson $\leftrightarrow$ Gamma
- Normal $\leftrightarrow$ Normal / Inverse-Gamma

4.11 Interpretation and Model Limitations

Remark 4.11.1. The Poisson model assumes:

- independence of events
- constant rate θ
- equal mean and variance

Remark 4.11.2. If variance exceeds the mean, a negative binomial model may be more appropriate.

Chapter 5

Monte Carlo Methods

5.1 Motivation

Remark 5.1.1. In many Bayesian models, posterior quantities cannot be computed analytically. Instead, we approximate expectations using simulation methods.

Definition 5.1.1. We are often interested in computing:

$$\mathbb{E}[f(\theta)] = \int f(\theta)p(\theta)\,d\theta$$

Remark 5.1.2. Monte Carlo methods approximate this integral using random samples.

5.2 Monte Carlo Estimation

Definition 5.2.1. Let $\theta^{(1)}, \ldots, \theta^{(S)} \sim p(\theta)$ be i.i.d. samples. Then:

$$\hat{\mu} = \frac{1}{S}\sum_{s=1}^{S} f(\theta^{(s)})$$

Proposition 5.2.1. *By the Law of Large Numbers:*

$$\hat{\mu} \to \mathbb{E}[f(\theta)] \quad \textit{as } S \to \infty$$

5.2.1 Variance and Error

Definition 5.2.2. Monte Carlo variance:

$$\widehat{\mathrm{Var}} = \frac{1}{S-1}\sum_{s=1}^{S}\left(f(\theta^{(s)}) - \hat{\mu}\right)^2$$

Definition 5.2.3. Monte Carlo standard error:

$$\text{MCSE} = \sqrt{\frac{\widehat{\text{Var}}}{S}}$$

Definition 5.2.4. Monte Carlo interval:

$$\hat{\mu} \pm 2 \cdot \text{MCSE}$$

If $\text{Var}\{f(\theta)\} < \infty$, the central limit theorem gives

$$\sqrt{S}\{\hat{\mu} - \mathbb{E}f(\theta)\} \xrightarrow{d} \mathcal{N}\Big(0, \text{Var}\{f(\theta)\}\Big).$$

The familiar Monte Carlo interval is therefore an interval for numerical error conditional on the fitted posterior; it is not a posterior credible interval for a model parameter.

Example 5.2.1 (Posterior tail probability)**.** After observing $y = 7$ successes in $n = 10$ trials under a uniform prior, $\theta \mid y \sim \text{Beta}(8, 4)$. To estimate $\mathbb{P}(\theta > 0.6 \mid y)$, draw $\theta^{(s)}$ from this beta posterior and use

$$\hat{p} = \frac{1}{S} \sum_{s=1}^{S} \mathbb{1}\{\theta^{(s)} > 0.6\}.$$

Because the summands are Bernoulli, an estimated numerical standard error is $\sqrt{\hat{p}(1-\hat{p})/S}$. If $\hat{p} = 0.72$ with $S = 10{,}000$, the MCSE is about 0.0045; simulation noise is then much smaller than the posterior uncertainty about θ.

5.3 Importance Sampling

Remark 5.3.1. Direct sampling from $p(\theta)$ is often difficult. Instead, we sample from an easier distribution $g(\theta)$.

Definition 5.3.1. Rewrite the expectation:

$$\mathbb{E}[f(\theta)] = \int f(\theta) \frac{p(\theta)}{g(\theta)} g(\theta)\, d\theta$$

Definition 5.3.2. Define weights:

$$w(\theta) = \frac{p(\theta)}{g(\theta)}$$

5.3.1 Estimators

- Unnormalized:

$$\hat{\mu}^* = \frac{1}{S} \sum_{s=1}^{S} f(\theta^{(s)}) w^*(\theta^{(s)})$$

- Normalized:

$$\hat{\mu} = \sum_{s=1}^{S} f(\theta^{(s)}) w(\theta^{(s)})$$

where:

$$w(\theta^{(s)}) = \frac{w^*(\theta^{(s)})}{\sum_{j=1}^{S} w^*(\theta^{(j)})}$$

5.3.2 Effective Sample Size

Definition 5.3.3. Effective sample size (ESS):

$$\text{ESS} \approx \frac{S}{1 + \text{Var}(w^*)}$$

Remark 5.3.2. High variance in weights reduces efficiency.

For normalized weights $\widetilde{w}_s = w_s / \sum_j w_j$, the commonly used realized diagnostic is

$$\widehat{\text{ESS}} = \frac{1}{\sum_{s=1}^{S} \widetilde{w}_s^2}.$$

It ranges from one, when a single draw carries all weight, to S, when the weights are equal. Finite variance requires the proposal to have sufficiently heavy tails: a proposal that undersamples an important target region can produce apparently stable estimates before a rare enormous weight appears.

Example 5.3.1 (Estimating a Gaussian tail)**.** Let $Z \sim \mathcal{N}(0,1)$ and seek $p = \mathbb{P}(Z > 5)$. Crude Monte Carlo sees almost no exceedances. Instead draw $X_s \sim \mathcal{N}(5,1)$ and compute

$$\hat{p} = \frac{1}{S} \sum_{s=1}^{S} \mathbb{1}\{X_s > 5\} \frac{\phi(X_s)}{\phi(X_s - 5)}.$$

The proposal deliberately places samples in the rare-event region, while the likelihood ratio corrects the change of measure. The example illustrates that a good proposal is tailored to the integrand as well as the target density.

5.4 Accept-Reject Sampling

Definition 5.4.1. Suppose we want to sample from $p(x)$ using proposal $g(x)$ such that:

$$p(x) \leq M g(x)$$

5.4.1 Algorithm

1. Sample $X \sim g(x)$
2. Sample $U \sim \text{Uniform}(0,1)$

3. Accept if:
$$U \leq \frac{p(X)}{Mg(X)}$$

4. Otherwise reject and repeat

Remark 5.4.1. Efficiency depends on choosing M as small as possible.

Conditional on a proposal $X = x$, acceptance occurs with probability $p(x)/(Mg(x))$. Hence

$$\mathbb{P}(X \in A,\ \text{accept}) = \int_A g(x)\frac{p(x)}{Mg(x)}\,\mathrm{d}x = \frac{1}{M}\int_A p(x)\,\mathrm{d}x.$$

Since the unconditional acceptance probability is $1/M$, division shows that an accepted proposal has density p. The expected number of proposals per accepted draw is M.

Example 5.4.1 (Beta target with a uniform envelope)**.** For $p(x) = 6x(1-x)$ on $[0,1]$ and $g(x) = 1$, the maximum of p/g is $M = 3/2$ at $x = 1/2$. Accept a uniform proposal X when $U \leq 4X(1-X)$. The acceptance rate is $2/3$, so the method needs an average of 1.5 proposals per beta draw.

5.5 Limitations of Direct Methods

Remark 5.5.1. Monte Carlo methods require:

- ability to sample from target distribution
- tractable density evaluation

Remark 5.5.2. When this is not possible, we use Markov Chain Monte Carlo (MCMC).

5.6 Markov Chains

Definition 5.6.1. A Markov chain satisfies:

$$P(X_{n+1} \mid X_n, \ldots, X_0) = P(X_{n+1} \mid X_n)$$

Definition 5.6.2. A stationary distribution π satisfies:

$$\pi = \pi P$$

Remark 5.6.1. We construct chains whose stationary distribution is the posterior $p(\theta|y)$.

5.7 Markov Chain Monte Carlo (MCMC)

Definition 5.7.1. MCMC methods generate dependent samples $\theta^{(1)}, \ldots, \theta^{(S)}$ such that:

$$\theta^{(s)} \sim p(\theta \mid y) \quad \text{as } s \to \infty$$

Proposition 5.7.1. *Ergodic theorem:*

$$\frac{1}{S}\sum_{s=1}^{S} f(\theta^{(s)}) \to \mathbb{E}[f(\theta \mid y)]$$

Remark 5.7.1. Samples are no longer independent, which affects variance.

5.8 Metropolis-Hastings Algorithm

Definition 5.8.1. Given proposal $q(\theta^* \mid \theta)$, define acceptance probability:

$$\alpha(\theta, \theta^*) = \min\left\{\frac{p(\theta^* \mid y)q(\theta \mid \theta^*)}{p(\theta \mid y)q(\theta^* \mid \theta)}, 1\right\}$$

5.8.1 Algorithm

1. Initialize $\theta^{(0)}$
2. Propose $\theta^* \sim q(\cdot \mid \theta^{(t-1)})$
3. Accept with probability α
4. Otherwise keep $\theta^{(t)} = \theta^{(t-1)}$

Remark 5.8.1. If q is symmetric, the ratio simplifies.

5.9 Random Walk Metropolis

Definition 5.9.1. Proposal:

$$\theta^* = \theta^{(t-1)} + \epsilon$$

Remark 5.9.1. Common choices for ϵ:

- Normal
- Student-t
- Uniform

5.10 Gibbs Sampling

Definition 5.10.1. Instead of sampling jointly, we sample from conditionals:

$$\theta_1 \sim p(\theta_1 \mid \theta_2, y), \quad \theta_2 \sim p(\theta_2 \mid \theta_1, y)$$

Remark 5.10.1. This is especially useful in high-dimensional problems.

5.11 Burn-in and Convergence

Definition 5.11.1. Burn-in: discard first B samples:

$$\{\theta^{(B+1)}, \ldots, \theta^{(B+S)}\}$$

Remark 5.11.1. Early samples depend on initialization and may not represent the target distribution.

5.12 Autocorrelation and Efficiency

Definition 5.12.1. Autocorrelation at lag t:

$$\rho_t = \frac{\sum_{s=1}^{S-t}(\theta_s - \bar{\theta})(\theta_{s+t} - \bar{\theta})}{\sum_{s=1}^{S}(\theta_s - \bar{\theta})^2}$$

Remark 5.12.1. High autocorrelation implies slow mixing and inefficient sampling.

5.13 Diagnostics

A trace plot should be examined for drift, sticking, slow movement between modes, and disagreement among independently initialized chains. Visual stability is necessary but not sufficient: a chain can look stationary while remaining trapped in one mode.

For m chains of retained length n, let W be the average within-chain variance and let B/n be the variance of the chain means. The original potential scale reduction factor estimates

$$\widehat{R} = sqrt\frac{(n-1)W/n + B/n}{W}.$$

Values appreciably above one indicate that between-chain dispersion remains larger than within-chain dispersion. Modern rank-normalized split-$\widehat{R}$ also detects scale and tail problems and should be reported together with bulk and tail effective sample sizes.

The Geweke statistic compares means from early and late windows using a spectral estimate of their Monte Carlo variances. It can reveal nonstationary behavior within a chain, but repeated testing and arbitrary window choices limit its interpretation. No finite collection of diagnostics proves convergence; diagnostics are attempts to falsify the claim that the simulation is reliable.

5.14 Summary

Independent Monte Carlo converts posterior integration into sampling error that can be quantified by a central limit theorem. Importance sampling and accept–reject sampling use tractable proposals in different ways: the former reweights every proposal, while the latter returns exact target draws by discarding proposals. MCMC is needed when neither direct construction is practical, but dependence replaces nominal sample size by an effective sample size and makes convergence assessment part of the analysis.

Chapter 6

Normal Model

6.1 The Normal Sampling Model

Definition 6.1.1. A random variable Y follows a normal distribution if:

$$Y \mid \theta, \sigma^2 \sim \mathcal{N}(\theta, \sigma^2)$$

with density:

$$p(y \mid \theta, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(y-\theta)^2\right)$$

Proposition 6.1.1. *For $Y \sim \mathcal{N}(\theta, \sigma^2)$:*

$$\mathbb{E}[Y] = \theta, \quad \mathrm{Var}(Y) = \sigma^2$$

Remark 6.1.1. The normal distribution is symmetric with mean = median = mode, and approximately 95% of mass lies within $\theta \pm 1.96\sigma$.

6.2 Likelihood Function

Let $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \mathcal{N}(\theta, \sigma^2)$. Then:

$$p(y \mid \theta, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i-\theta)^2\right)$$

Remark 6.2.1. Inference can be separated into:

- inference on θ given σ^2
- inference on σ^2

6.3 Known Variance: Conjugate Analysis

6.3.1 Prior

Definition 6.3.1. Assume:

$$\theta \sim \mathcal{N}(\mu_0, \tau_0^2)$$

6.3.2 Posterior

Theorem 6.3.1. *If* $Y_i \overset{iid}{\sim} \mathcal{N}(\theta, \sigma^2)$ *and* $\theta \sim \mathcal{N}(\mu_0, \tau_0^2)$*, then:*

$$\theta \mid y \sim \mathcal{N}(\mu_n, \tau_n^2)$$

where:

$$\frac{1}{\tau_n^2} = \frac{1}{\tau_0^2} + \frac{n}{\sigma^2}$$

$$\mu_n = \frac{\frac{1}{\tau_0^2}\mu_0 + \frac{n}{\sigma^2}\bar{y}}{\frac{1}{\tau_0^2} + \frac{n}{\sigma^2}}$$

Proof. Posterior:

$$p(\theta \mid y) \propto p(\theta) \cdot p(y \mid \theta)$$

Combine exponent terms:

$$-\frac{1}{2\tau_0^2}(\theta - \mu_0)^2 - \frac{1}{2\sigma^2}\sum(y_i - \theta)^2$$

Expand and collect quadratic terms in θ, yielding a normal kernel. □

6.3.3 Interpretation

Remark 6.3.1. The posterior mean is a weighted average:

$$\mu_n = \frac{\kappa_0}{\kappa_0 + n}\mu_0 + \frac{n}{\kappa_0 + n}\bar{y}$$

where $\kappa_0 = \sigma^2/\tau_0^2$.

Remark 6.3.2. Precision (inverse variance) is additive:

$$\text{precision}_{posterior} = \text{precision}_{prior} + \text{precision}_{data}$$

6.4 Unknown Variance: Joint Inference

Remark 6.4.1. When σ^2 is unknown, we introduce a prior on variance.

6.4.1 Prior Specification

Definition 6.4.1. Use:

$$\theta \mid \sigma^2 \sim \mathcal{N}(\mu_0, \sigma^2/\kappa_0)$$
$$\sigma^2 \sim \text{Inv-Gamma}\left(\frac{\nu_0}{2}, \frac{\nu_0\sigma_0^2}{2}\right)$$

6.4.2 Posterior Updates

Proposition 6.4.1. *Posterior parameters:*

$$\kappa_n = \kappa_0 + n$$
$$\mu_n = \frac{\kappa_0\mu_0 + n\bar{y}}{\kappa_n}\nu_n = \nu_0 + n$$
$$\sigma_n^2 = \frac{1}{\nu_n}\left[\nu_0\sigma_0^2 + (n-1)s^2 + \frac{\kappa_0 n}{\kappa_n}(\bar{y} - \mu_0)^2\right]$$

Remark 6.4.2. This is often summarized as: "posterior sum of squares = prior sum + data sum + correction term."

6.5 Conditional Posteriors

Proposition 6.5.1.

$$\theta \mid \sigma^2, y \sim \mathcal{N}(\mu_n, \sigma^2/\kappa_n)$$
$$\sigma^2 \mid y \sim \text{Inv-Gamma}\left(\frac{\nu_n}{2}, \frac{\nu_n\sigma_n^2}{2}\right)$$

6.6 Marginal Posterior for the Mean

Theorem 6.6.1. *The marginal posterior of θ is:*

$$\frac{\theta - \mu_n}{\sigma_n/\sqrt{\kappa_n}} \sim t_{\nu_n}$$

Remark 6.6.1. The t-distribution arises from integrating out uncertainty in σ^2.

6.7 Example: Midge Wing Length

Example 6.7.1. Data:

$$y = (1.64, 1.70, 1.72, 1.74, 1.82, 1.82, 1.82, 1.90, 2.08)$$

Summary:

$$\bar{y} = 1.804, \quad s^2 \approx 0.017$$

Using prior $\mu_0 = 1.9$, $\kappa_0 = 1$, we obtain:

$$\mu_n \approx 1.814, \quad \sigma_n^2 \approx 0.015$$

6.8 Sampling from the Posterior

The factorization

$$p(\theta, \sigma^2 \mid y) = p(\theta \mid \sigma^2, y)p(\sigma^2 \mid y)$$

provides independent joint posterior draws: first sample $\sigma^{2(s)}$ from its inverse-gamma marginal, then sample $\theta^{(s)}$ from the conditional normal. These are not MCMC draws, because each pair is generated independently of previous pairs.

The posterior predictive distribution for one new observation is

$$\frac{\widetilde{Y} - \mu_n}{\sigma_n \sqrt{1 + 1/\kappa_n}} \sim t_{\nu_n}.$$

The additional 1 inside the scale accounts for future sampling noise; the $1/\kappa_n$ term accounts for uncertainty about the mean.

Example 6.8.1 (Prediction versus parameter inference)**.** If $\sigma_n = 2$ and $\kappa_n = 25$, the posterior standard scale for the mean is $2/5 = 0.4$, whereas the predictive scale is $2\sqrt{1 + 1/25} \approx 2.04$. Predicting an individual is much less precise than estimating the population mean because individuals retain observation noise even when the mean is well learned.

6.9 Bias and Mean Squared Error

Definition 6.9.1. Bias:

$$\mathrm{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$$

Definition 6.9.2. Mean Squared Error:

$$\mathrm{MSE}(\hat{\theta}) = \mathrm{Var}(\hat{\theta}) + \mathrm{Bias}^2$$

Remark 6.9.1. Bayesian estimators are typically biased but can have lower MSE due to shrinkage.

6.10 Exponential Family Representation

Proposition 6.10.1. *The normal distribution belongs to the exponential family:*

$$p(y \mid \theta, \sigma^2) = \exp\left(\frac{\theta}{\sigma^2}y - \frac{1}{2\sigma^2}y^2 - A(\theta, \sigma^2)\right)$$

Remark 6.10.1. Sufficient statistics:

$$t(y) = (y, y^2)$$

6.11 Summary

Normal conjugacy is precision-weighted updating. When variance is unknown, integrating it out produces Student distributions with heavier tails for both the mean and future observations. The distinction between a posterior interval for the population mean and a predictive interval for an individual is essential: the latter contains irreducible sampling variation.

Chapter 7

Multinomial Model, Bayes Factors, and Indirect Methods

7.1 Multinomial Model

7.1.1 Model Setup

Definition 7.1.1. Let:

$$x = (x_1, \ldots, x_k), \quad \sum_{j=1}^{k} x_j = n$$

Then the multinomial model is:

$$p(x \mid \theta) = \frac{n!}{\prod_{j=1}^{k} x_j!} \prod_{j=1}^{k} \theta_j^{x_j}$$

where:

$$\theta = (\theta_1, \ldots, \theta_k), \quad \sum_{j=1}^{k} \theta_j = 1$$

Remark 7.1.1. This generalizes the binomial model to k categories.

7.1.2 Dirichlet Prior

Definition 7.1.2. The conjugate prior is:

$$\theta \sim \text{Dirichlet}(\alpha)$$

with density:

$$\pi(\theta) = \frac{1}{B(\alpha)} \prod_{j=1}^{k} \theta_j^{\alpha_j - 1}$$

Remark 7.1.2.

- α_j acts like prior counts
- $\alpha_j = 1$ gives a uniform prior
- when $k = 2$, Dirichlet reduces to Beta

7.1.3 Posterior

Theorem 7.1.1. *Given data x, the posterior is:*

$$\theta \mid x \sim \text{Dirichlet}(\alpha + x)$$

Proof.

$$p(\theta \mid x) \propto \prod_{j=1}^{k} \theta_j^{x_j} \cdot \prod_{j=1}^{k} \theta_j^{\alpha_j - 1} = \prod_{j=1}^{k} \theta_j^{x_j + \alpha_j - 1}$$

□

7.1.4 Interpretation

Let $\alpha_0 = \sum_j \alpha_j$. The prior moments are

$$\mathbb{E}(\theta_j) = \frac{\alpha_j}{\alpha_0}, \quad \text{Var}(\theta_j) = \frac{\alpha_j(\alpha_0 - \alpha_j)}{\alpha_0^2(\alpha_0 + 1)},$$

and, for $j \neq \ell$,

$$\text{Cov}(\theta_j, \theta_\ell) = -\frac{\alpha_j \alpha_\ell}{\alpha_0^2(\alpha_0 + 1)}.$$

The negative covariance is forced by the simplex constraint. Posterior updating replaces α_j by $\alpha_j + x_j$, so the posterior predictive probability of the next category is

$$\mathbb{P}(\widetilde{Y} = j \mid x) = \mathbb{E}(\theta_j \mid x) = \frac{\alpha_j + x_j}{\alpha_0 + n}.$$

This is additive smoothing: categories with zero observed counts retain positive predictive probability when $\alpha_j > 0$.

Example 7.1.1 (Three-category survey)**.** A survey records counts $(50, 30, 20)$ across three responses. Under Dirichlet$(1, 1, 1)$, the posterior is Dirichlet$(51, 31, 21)$ and the next-response probabilities are $(51, 31, 21)/103$. The unobserved future category is random even after fixing θ, while uncertainty about θ creates additional dependence among multiple future responses. Integrating θ gives the Dirichlet–multinomial predictive distribution.

7.2 Bayes Factors

7.2.1 Definition

Definition 7.2.1. Given two models M_1 and M_2:

$$\text{BF} = \frac{p(M_1 \mid x)}{p(M_2 \mid x)} \cdot \frac{p(M_2)}{p(M_1)} = \frac{p(x \mid M_1)}{p(x \mid M_2)}$$

Remark 7.2.1. Bayes factor compares how well each model explains the data.

7.2.2 Marginal Likelihood

Definition 7.2.2.

$$p(x \mid M) = \int p(x \mid \theta, M)\pi(\theta \mid M)\, d\theta$$

Remark 7.2.2. This integrates out parameter uncertainty.

7.2.3 Interpretation

Remark 7.2.3. Typical interpretation scales (Jeffreys):

- $1 \leq BF < 3$: weak evidence
- $3 \leq BF < 10$: substantial
- $10 \leq BF < 100$: strong
- $BF \geq 100$: decisive

7.2.4 Important Caveats

Remark 7.2.4.

- Bayes factors require **proper priors**
- If prior is improper $\Rightarrow$ marginal likelihood undefined
- Sensitive to prior specification

7.2.5 Frequentist vs Bayesian Testing

Remark 7.2.5.

- Frequentist: reject/not reject (no strength)
- Bayesian: provides strength of evidence via BF

7.3 Indirect Methods (Monte Carlo)

7.3.1 Motivation

Remark 7.3.1. We often need to compute:

$$\mathbb{E}[f(\theta)] = \int f(\theta)p(\theta)\, d\theta$$

but cannot evaluate the integral analytically.

7.4 Importance Sampling

7.4.1 Core Idea

Definition 7.4.1. Rewrite expectation using proposal $g(x)$:

$$\mathbb{E}[h(x)] = \int h(x)f(x)\,dx = \int h(x)\frac{f(x)}{g(x)}g(x)\,dx$$

7.4.2 Estimator

Definition 7.4.2. Unnormalized:

$$\hat{\mu}^* = \frac{1}{n}\sum_{i=1}^{n} h(X_i)\frac{f(X_i)}{g(X_i)}$$

Normalized:

$$\hat{\mu} = \sum_{i=1}^{n} h(X_i)w_i$$

where:

$$w_i = \frac{f(X_i)/g(X_i)}{\sum_j f(X_j)/g(X_j)}$$

7.4.3 Effective Sample Size

Definition 7.4.3.

$$\text{ESS} = \frac{n}{1 + \widehat{\text{Var}}(w^*)}$$

Remark 7.4.1. High variance weights $\Rightarrow$ low efficiency.

7.5 Accept-Reject Sampling

7.5.1 Algorithm

7.5.2 Remarks

- Smaller $M \Rightarrow$ higher efficiency
- g must be easy to sample

7.6 Monte Carlo Approximation

Definition 7.6.1. Given samples $\theta^{(1)}, \ldots, \theta^{(S)}$:

$$\hat{\theta} = \frac{1}{S}\sum_{s=1}^{S} \theta^{(s)}$$

1. Choose proposal $g(x)$ and constant M such that:

$$f(x) \leq Mg(x)$$

2. Sample $X \sim g(x)$
3. Sample $U \sim \text{Uniform}(0, 1)$
4. Accept if:

$$U \leq \frac{f(X)}{Mg(X)}$$

Definition 7.6.2. Monte Carlo variance:

$$\widehat{\text{Var}} = \frac{1}{S-1} \sum (\theta^{(s)} - \bar{\theta})^2$$

Remark 7.6.1. Accuracy improves with S by law of large numbers.

7.7 Summary

- Multinomial + Dirichlet gives clean conjugate updates
- Bayes factors compare models via marginal likelihoods
- Indirect methods approximate integrals when closed forms fail
- Importance sampling and AR are foundational tools before MCMC

Chapter 8

Markov Chain Monte Carlo, Gibbs Sampling, and Diagnostics

8.1 Motivation

Remark 8.1.1. In many Bayesian models, the posterior

$$p(\theta \mid y)$$

cannot be computed in closed form.

We instead approximate expectations:

$$\mathbb{E}[h(\theta) \mid y] = \int h(\theta) p(\theta \mid y)\, d\theta$$

using simulation methods.

8.2 Markov Chains

8.2.1 Definition

Definition 8.2.1. A sequence $\{X_t\}$ is a Markov chain if:

$$\mathbb{P}(X_{t+1} \mid X_t, \ldots, X_0) = \mathbb{P}(X_{t+1} \mid X_t)$$

8.2.2 Key Properties

Irreducibility means that every set of positive target probability is reachable from every admissible starting point. Aperiodicity rules out a deterministic cycle of return times. Positive recurrence ensures finite mean return times and the existence of a normalizable stationary distribution. In a general state space these conditions are expressed through π-irreducibility and Harris recurrence. Together with mild regularity, they justify convergence to stationarity and the Markov-chain ergodic theorem.

8.2.3 Stationary Distribution

Definition 8.2.2. A distribution π is stationary if:

$$\pi = \pi P$$

Remark 8.2.1. MCMC constructs a Markov chain whose stationary distribution is the posterior:

$$\pi(\theta) = p(\theta \mid y)$$

8.3 Markov Chain Monte Carlo

Definition 8.3.1. MCMC generates dependent samples:

$$\theta^{(1)}, \theta^{(2)}, \dots$$

such that:

$$\theta^{(t)} \xrightarrow{d} p(\theta \mid y)$$

Proposition 8.3.1. *For an ergodic chain:*

$$\mathbb{E}[h(\theta) \mid y] \approx \frac{1}{T} \sum_{t=1}^{T} h(\theta^{(t)})$$

8.4 Metropolis-Hastings Algorithm

Definition 8.4.1. Given a proposal $q(\theta^* \mid \theta)$, define:

$$\alpha(\theta, \theta^*) = \min \left\{ \frac{p(\theta^* \mid y) q(\theta \mid \theta^*)}{p(\theta \mid y) q(\theta^* \mid \theta)}, 1 \right\}$$

---

1. Initialize $\theta^{(0)}$
2. For $t = 1, \dots, T$:
   (a) Sample $\theta^* \sim q(\cdot \mid \theta^{(t-1)})$
   (b) Accept with probability α
   (c) Otherwise set $\theta^{(t)} = \theta^{(t-1)}$

Remark 8.4.1. If q is symmetric, the ratio simplifies to:

$$\alpha = \min \left\{ \frac{p(\theta^* \mid y)}{p(\theta \mid y)}, 1 \right\}$$

Proposition 8.4.1. *The Metropolis–Hastings transition is reversible with respect to the target π and therefore leaves π invariant.*

Proof. For distinct x and y, the accepted-move density satisfies

$$\begin{aligned}\pi(x)q(y \mid x)\alpha(x,y) &= \min\{\pi(x)q(y \mid x), \pi(y)q(x \mid y)\} \\ &= \pi(y)q(x \mid y)\alpha(y,x).\end{aligned}$$

The rejection probability contributes only to the diagonal and also satisfies detailed balance. Integrating over the current state proves that π is stationary. □

Example 8.4.1 (Independence sampler for a beta posterior)**.** Suppose π is a Beta$(8,4)$ posterior and proposals are drawn from $q = \text{Beta}(4,4)$ independently of the current state. Then

$$\alpha(\theta, \theta^*) = \min\left\{1, \frac{\pi(\theta^*)q(\theta)}{\pi(\theta)q(\theta^*)}\right\}.$$

Normalizing constants cancel within each density ratio. Unlike a random walk, this proposal can make global moves, but it performs poorly if q is too light-tailed or misses an important posterior region.

8.5 Random Walk Metropolis

Definition 8.5.1. A common proposal is:

$$\theta^* = \theta^{(t-1)} + \epsilon$$

Remark 8.5.1. The choice of proposal variance determines:

- step size
- acceptance rate
- efficiency

8.6 Gibbs Sampling

8.6.1 Definition

Definition 8.6.1. Let $\theta = (\theta_1, \ldots, \theta_k)$. Gibbs sampling iteratively samples:

$$\begin{aligned}\theta_1^{(t)} &\sim p(\theta_1 \mid \theta_2^{(t-1)}, \ldots, \theta_k^{(t-1)}, y) \\ \theta_2^{(t)} &\sim p(\theta_2 \mid \theta_1^{(t)}, \ldots, \theta_k^{(t-1)}, y) \\ &\vdots \\ \theta_k^{(t)} &\sim p(\theta_k \mid \theta_1^{(t)}, \ldots, \theta_{k-1}^{(t)}, y)\end{aligned}$$

Remark 8.6.1. Gibbs sampling avoids rejection by sampling directly from full conditional distributions.

8.6.2 Example: Normal Model

For $Y_i \mid \mu, \sigma^2 \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$, take $\mu \mid \sigma^2 \sim \mathcal{N}(\mu_0, \sigma^2/\kappa_0)$ and an inverse-gamma prior $\sigma^2 \sim$ Inv-Gamma(a_0, b_0). Completing the square gives

$$\mu \mid \sigma^2, y \sim \mathcal{N}\left(\mu_n, \frac{\sigma^2}{\kappa_n}\right), \quad \kappa_n = \kappa_0 + n, \quad \mu_n = \frac{\kappa_0 \mu_0 + n\bar{y}}{\kappa_n}.$$

Conditional on a current value of μ,

$$\sigma^2 \mid \mu, y \sim \text{Inv-Gamma}\left(a_0 + \frac{n+1}{2}, b_0 + \frac{\sum_i (y_i - \mu)^2 + \kappa_0(\mu - \mu_0)^2}{2}\right).$$

Alternating these distributions yields a Gibbs chain. Because this model is conjugate, direct joint or collapsed sampling is also possible and generally has lower autocorrelation; Gibbs sampling is pedagogically useful here rather than computationally necessary.

8.7 Burn-in

Definition 8.7.1. Discard the first B samples:

$$\{\theta^{(B+1)}, \ldots, \theta^{(T)}\}$$

Remark 8.7.1. Initial samples depend on the starting value and may not represent the posterior.

8.8 Autocorrelation

Definition 8.8.1. Lag-k autocorrelation:

$$\rho_k = \frac{\sum_{t=1}^{T-k} (\theta^{(t)} - \bar{\theta})(\theta^{(t+k)} - \bar{\theta})}{\sum_{t=1}^{T} (\theta^{(t)} - \bar{\theta})^2}$$

Remark 8.8.1. High autocorrelation implies slow exploration of the parameter space.

8.9 Effective Sample Size

Definition 8.9.1.

$$\text{ESS} = \frac{T}{1 + 2\sum_{k=1}^{\infty} \rho_k}$$

Remark 8.9.1. ESS measures the number of independent samples equivalent to the correlated chain.

Example 8.9.1 (AR(1) benchmark)**.** If a stationary scalar chain has $\rho_k = \rho^k$ with $0 < \rho < 1$, then

$$\tau_{\text{int}} = 1 + 2\sum_{k\geq 1} \rho^k = \frac{1+\rho}{1-\rho}.$$

With $T = 10{,}000$ and $\rho = 0.9$, the ESS is only $T/\tau_{\text{int}} \approx 526$. Saving every tenth draw does not create information; it mostly discards correlated draws that could still improve an estimate.

8.10 Trace Plots

Remark 8.10.1. Plotting $\theta^{(t)}$ against t provides a visual diagnostic:

- stable band indicates convergence
- trends or clustering indicate poor mixing

8.11 Formal Diagnostics

Run multiple chains from deliberately dispersed initial states and split each chain before computing $\widehat{R}$, since disagreement between early and late halves reveals nonstationarity. Rank normalization makes the diagnostic less sensitive to heavy tails, while folding draws around their median helps detect scale differences. A practical analysis reports split $\widehat{R}$, bulk ESS for means, and tail ESS for quantiles.

Geweke's statistic compares early- and late-window means using spectral variance estimates. Heidelberger–Welch combines a stationarity test with a half-width calculation, while Raftery–Lewis estimates the run length needed for a chosen quantile precision. These procedures address different failure modes. Passing them does not prove that an isolated mode was found or that all posterior functionals are accurately estimated.

8.12 Mixing

Definition 8.12.1. Mixing describes how quickly the chain explores the parameter space.

Mixing is a property of the full transition kernel, not just one plotted coordinate. A chain can have low autocorrelation within a mode and still cross modes extremely rarely. Strong posterior correlations create narrow ridges, incompatible parameter scales make one random-walk variance inadequate, and hierarchical funnels couple local and global scales.

8.13 Improving MCMC Performance

Running longer reduces error only after the chain can explore the relevant posterior regions. Proposal tuning balances movement size against acceptance, and covariance adaptation aligns proposals with posterior scale. Blocking correlated variables, marginalizing nuisance

parameters, or using a noncentered hierarchical parameterization can change mixing by orders of magnitude. Gradient-based methods such as Hamiltonian Monte Carlo avoid much of the high-dimensional random-walk behavior by following approximate posterior level sets.

8.14 Rao–Blackwellization and Collapsing

If (Θ, Z) is sampled jointly and the conditional expectation $m(Z) = \mathbb{E}\{h(\Theta) \mid Z, y\}$ is available, then

$$\mathbb{E}\{m(Z) \mid y\} = \mathbb{E}\{h(\Theta) \mid y\}$$

and the law of total variance gives

$$\operatorname{Var}\{h(\Theta) \mid y\} = \mathbb{E}[\operatorname{Var}\{h(\Theta) \mid Z, y\} \mid y] + \operatorname{Var}\{m(Z) \mid y\}.$$

Therefore averaging $m(Z^{(t)})$ has no larger marginal variance than averaging $h(\Theta^{(t)})$. In a Markov chain the autocovariances also matter, so the exact efficiency gain depends on the transition, but conditional averaging often reduces Monte Carlo noise substantially.

Collapsing goes further by integrating a variable out of the target before constructing the transition. In a Dirichlet–multinomial mixture, integrating out component probabilities creates a sampler on allocations alone. The collapsed chain has lower dimension and can mix faster, although deriving its conditional probabilities requires more algebra and careless partial collapsing can destroy stationarity.

8.15 Hamiltonian Monte Carlo

For a continuous parameter with target $\pi(q) \propto e^{-U(q)}$, Hamiltonian Monte Carlo introduces momentum $p \sim \mathcal{N}(0, M)$ and Hamiltonian

$$H(q, p) = U(q) + \frac{1}{2} p^{\top} M^{-1} p.$$

Hamilton's equations are

$$\frac{\mathrm{d}q}{\mathrm{d}t} = M^{-1} p, \qquad \frac{\mathrm{d}p}{\mathrm{d}t} = -\nabla U(q).$$

Their exact flow conserves H, preserves volume, and is reversible after a momentum flip. Consequently it can move a long distance across the posterior without the random-walk behavior of a local proposal.

The leapfrog integrator alternates

$$\begin{aligned} p_{\ell+1/2} &= p_{\ell} - \frac{\epsilon}{2} \nabla U(q_{\ell}), \\ q_{\ell+1} &= q_{\ell} + \epsilon M^{-1} p_{\ell+1/2}, \\ p_{\ell+1} &= p_{\ell+1/2} - \frac{\epsilon}{2} \nabla U(q_{\ell+1}). \end{aligned}$$

Because numerical integration introduces energy error, the endpoint is accepted with probability $\min\{1, \exp[-H(q^*, p^*) + H(q, p)]\}$. Leapfrog reversibility and volume preservation make this correction valid.

Example 8.15.1 (Gaussian target)**.** For a standard normal target, $U(q) = q^2/2$ and the exact Hamiltonian dynamics satisfy $q''(t) = -q(t)$. Hence

$$q(t) = q(0)\cos t + p(0)\sin t.$$

The trajectory moves across the distribution on an ellipse while preserving $q^2 + p^2$. Random-walk Metropolis instead diffuses across the same region one local step at a time.

The step size ϵ controls numerical error, the mass matrix M rescales correlated parameters, and the trajectory length controls how far a proposal travels. The no-U-turn sampler adapts trajectory length during warmup. Divergent transitions signal that the integrator encountered severe curvature; they should prompt reparameterization or stronger modeling choices rather than being hidden by a higher acceptance target.

8.16 Monte Carlo Error for MCMC

Under a Markov-chain central limit theorem,

$$\sqrt{T}\{\bar{h}_T - \mathbb{E}_\pi h\} \xrightarrow{d} \mathcal{N}(0, \sigma_h^2), \qquad \sigma_h^2 = \gamma_0 + 2\sum_{k=1}^{\infty} \gamma_k,$$

where $\gamma_k = \mathrm{Cov}_\pi\{h(\Theta_0), h(\Theta_k)\}$. The numerical standard error is $\sqrt{\hat{\sigma}_h^2/T}$, estimated using batch means or a spectral variance estimator. This quantity should be reported on the scale of the posterior summary. A large ESS for one coefficient does not imply a large ESS for a tail probability or nonlinear prediction.

Example 8.16.1 (Stopping by precision)**.** Suppose the posterior mean of a treatment effect is estimated as 1.24 and a batch-means calculation gives MCSE 0.01. If two decimal places are required, the simulation error is material and the chain should be extended. If the posterior standard deviation is 0.60, however, this numerical error is small relative to inferential uncertainty. The two scales answer different questions and should not be conflated.

8.17 Summary

Metropolis–Hastings obtains invariance through detailed balance, while Gibbs sampling exploits tractable full conditionals. Invariance is only the starting point: initialization, multimodality, posterior geometry, and autocorrelation determine whether a finite run is useful. Reliable reporting pairs multiple-chain diagnostics with effective sample sizes and checks the specific posterior quantities used in the final inference.

Chapter 9

Multivariate Normal Model

9.1 Multivariate Normal Distribution

Definition 9.1.1. A random vector $Y \in \mathbb{R}^p$ follows a multivariate normal distribution if:

$$Y \mid \theta, \Sigma \sim \mathcal{N}_p(\theta, \Sigma)$$

with density:

$$p(y \mid \theta, \Sigma) = \frac{1}{(2\pi)^{p/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(y-\theta)^\top \Sigma^{-1}(y-\theta)\right)$$

Proposition 9.1.1.

$$\mathbb{E}[Y] = \theta, \quad \mathrm{Var}(Y) = \Sigma$$

Remark 9.1.1. The covariance matrix Σ must be symmetric positive definite.

9.2 Marginal and Conditional Gaussian Laws

Partition a Gaussian vector and its moments as

$$Y = \begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix}, \qquad \mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \qquad \Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}.$$

Then $Y_1 \sim \mathcal{N}(\mu_1, \Sigma_{11})$, and completing the quadratic form shows

$$Y_1 \mid Y_2 = y_2 \sim \mathcal{N}\Big(\mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(y_2 - \mu_2), \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}\Big).$$

The conditional covariance is the Schur complement of Σ_{22} and is no larger than Σ_{11} in positive-semidefinite order. Observing correlated coordinates therefore reduces uncertainty.

Example 9.2.1 (Bivariate conditional prediction)**.** Suppose standardized systolic and diastolic blood pressure are jointly normal with correlation 0.8. Conditional on a systolic value two standard deviations above its mean, the conditional mean of standardized diastolic pressure is $0.8(2) = 1.6$, and its conditional variance is $1 - 0.8^2 = 0.36$. Marginal prediction would ignore both the shift and the reduction in variance.

The same formula underlies multivariate regression calculations and the imputation distribution used for multivariate normal missing data.

9.3 Sampling Model

Let $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \mathcal{N}_p(\theta, \Sigma)$.

Proposition 9.3.1. *The likelihood is:*

$$p(y \mid \theta, \Sigma) \propto |\Sigma|^{-n/2} \exp\left(-\frac{1}{2}\sum_{i=1}^{n}(y_i - \theta)^\top \Sigma^{-1}(y_i - \theta)\right)$$

Remark 9.3.1. The sufficient statistics are:

$$\bar{y} = \frac{1}{n}\sum y_i, \quad S = \sum (y_i - \bar{y})(y_i - \bar{y})^\top$$

9.4 Known Covariance Matrix

9.4.1 Prior

Definition 9.4.1.

$$\theta \sim \mathcal{N}_p(\mu_0, \Lambda_0)$$

9.4.2 Posterior

Theorem 9.4.1.

$$\theta \mid y, \Sigma \sim \mathcal{N}_p(\mu_n, \Lambda_n)$$

where:

$$\Lambda_n = \left(\Lambda_0^{-1} + n\Sigma^{-1}\right)^{-1}$$
$$\mu_n = \Lambda_n\left(\Lambda_0^{-1}\mu_0 + n\Sigma^{-1}\bar{y}\right)$$

Remark 9.4.1. Posterior precision:

$$\Lambda_n^{-1} = \Lambda_0^{-1} + n\Sigma^{-1}$$

Example 9.4.1 (Correlated prior and likelihood)**.** Let $p = 2$, $\Sigma = I$, $n = 4$, and $\Lambda_0 = \text{diag}(1, 1/4)$. Then

$$\Lambda_n = \text{diag}(1/5, 1/8).$$

The second coordinate has smaller posterior variance because its prior was four times as precise. If $\mu_0 = 0$ and $\bar{y} = (1, 1)^\top$, the posterior mean is $(4/5, 1/2)^\top$: the same data are shrunk differently in the two directions. In a non-diagonal problem the eigenvectors of the precision matrices determine the shrinkage directions.

Remark 9.4.2. Posterior mean is a weighted average of prior mean and sample mean.

9.5 Unknown Covariance Matrix

9.5.1 Inverse-Wishart Prior

Definition 9.5.1.

$$\Sigma \sim \text{Inv-Wishart}(\nu_0, S_0)$$

Remark 9.5.1. This is the multivariate analogue of the inverse-gamma distribution.

The inverse-Wishart couples marginal variances and correlations through one matrix parameter. This makes it conjugate but can make prior specification opaque: a choice intended to be weak for variances may still concentrate correlations. Prior predictive simulation of covariance matrices is therefore more informative than inspecting ν_0 and S_0 in isolation.

For interpretable modeling, decompose

$$\Sigma = DRD, \qquad D = \text{diag}(\sigma_1, \ldots, \sigma_p),$$

where R is a correlation matrix. Separate priors on scales and correlations can express knowledge more directly, though they sacrifice elementary inverse-Wishart conjugacy.

Example 9.5.1 (Correlation affects joint uncertainty)**.** For

$$\Sigma = \begin{pmatrix} 4 & 1.8 \\ 1.8 & 1 \end{pmatrix},$$

the standard deviations are 2 and 1, and the correlation is 0.9. Marginal intervals describe each coordinate separately, but the joint credible ellipse is narrow along one rotated direction and wide along the other. Reporting only marginal variances hides this geometry.

9.5.2 Joint Prior

Definition 9.5.2.

$$\theta \mid \Sigma \sim \mathcal{N}_p(\mu_0, \Sigma/\kappa_0)$$
$$\Sigma \sim \text{Inv-Wishart}(\nu_0, S_0)$$

9.6 Posterior Distribution

Proposition 9.6.1. *Posterior parameters:*

$$\kappa_n = \kappa_0 + n$$
$$\mu_n = \frac{\kappa_0 \mu_0 + n\bar{y}}{\kappa_n}$$
$$\nu_n = \nu_0 + n$$
$$S_n = S_0 + \sum_{i=1}^{n} (y_i - \bar{y})(y_i - \bar{y})^\top + \frac{\kappa_0 n}{\kappa_n} (\bar{y} - \mu_0)(\bar{y} - \mu_0)^\top$$

Proposition 9.6.2.

$$\theta \mid \Sigma, y \sim \mathcal{N}_p(\mu_n, \Sigma/\kappa_n)$$
$$\Sigma \mid y \sim \textit{Inv-Wishart}(\nu_n, S_n)$$

9.7 Gibbs Sampling for MVN Model

---

1. Initialize $\theta^{(0)}, \Sigma^{(0)}$
2. For $t = 1, \ldots, T$:
   (a) Sample:
   $$\theta^{(t)} \sim \mathcal{N}_p(\mu_n, \Sigma^{(t-1)}/\kappa_n)$$
   (b) Sample:
   $$\Sigma^{(t)} \sim \text{Inv-Wishart}(\nu_n, S_n(\theta^{(t)}))$$

Remark 9.7.1. This alternates between conditional distributions, producing samples from the joint posterior.

9.8 Wishart Distribution

Definition 9.8.1. The Wishart distribution is:

$$W \sim \text{Wishart}_p(\nu, \Sigma)$$

Proposition 9.8.1.

$$\mathbb{E}[W] = \nu\Sigma$$

Remark 9.8.1. The Wishart distribution is the distribution of a sample covariance matrix.

9.9 Inverse-Wishart Distribution

Proposition 9.9.1. *If* $\Sigma \sim \textit{Inv-Wishart}(\nu, S)$*:*

$$\mathbb{E}[\Sigma] = \frac{S}{\nu - p - 1}$$

Remark 9.9.1. Defined only when $\nu > p + 1$.

9.10 Multivariate t-Distribution

Proposition 9.10.1. *Marginalizing Σ yields:*

$$\theta \mid y \sim t_{\nu_n - p + 1}(\mu_n, S_n)$$

Remark 9.10.1. The multivariate t distribution accounts for uncertainty in covariance estimation.

9.11 Simulation of MVN

Proposition 9.11.1. *If $Z \sim \mathcal{N}_p(0, I)$ and $\Sigma = LL^\top$, then:*

$$Y = \theta + LZ \sim \mathcal{N}_p(\theta, \Sigma)$$

9.12 Summary

- Multivariate normal generalizes the normal model to vector-valued data
- Mean becomes a vector, variance becomes a covariance matrix
- Conjugate prior: Normal-Inverse-Wishart
- Posterior updates mirror univariate case using matrix algebra
- Gibbs sampling alternates between mean and covariance updates

Chapter 10

Group Comparisons and Hierarchical Models

10.1 Motivation

Grouped data contain two sources of variation. Students in the same school or patients in the same hospital may resemble one another, while group means also vary across the population. Estimating every group independently is unstable for small groups, whereas complete pooling erases genuine heterogeneity. A hierarchical model learns a compromise from the data.

10.2 Two-Group Comparison Model

10.2.1 Model Specification

Definition 10.2.1. For two groups:

$$Y_{i,1} = \mu + \delta + \epsilon_{i,1}$$
$$Y_{i,2} = \mu - \delta + \epsilon_{i,2}$$
$$\epsilon_{i,j} \sim \mathcal{N}(0, \sigma^2)$$

Remark 10.2.1. The two group means are $\mu + \delta$ and $\mu - \delta$. Thus μ is their midpoint, while the actual group-1 minus group-2 contrast is 2δ, not δ. The parameterization makes the location and contrast orthogonal in a balanced design; σ^2 is the shared within-group variance.

10.3 Priors

Definition 10.3.1.

$$\mu \sim \mathcal{N}(\mu_0, \gamma_0^2)$$
$$\delta \sim \mathcal{N}(\delta_0, \tau_0^2)$$
$$\sigma^2 \sim \text{Inv-Gamma}(\nu_0/2, \nu_0\sigma_0^2/2)$$

10.4 Posterior Conditionals

10.4.1 Posterior of μ

Proposition 10.4.1.

$$\mu \mid \delta, \sigma^2, y \sim \mathcal{N}(\mu_n, \gamma_n^2)$$

$$\gamma_n^2 = \left(\frac{1}{\gamma_0^2} + \frac{n_1 + n_2}{\sigma^2} \right)^{-1}$$

$$\mu_n = \gamma_n^2 \left[\frac{\mu_0}{\gamma_0^2} + \frac{\sum (y_{i,1} - \delta)}{\sigma^2} + \frac{\sum (y_{i,2} + \delta)}{\sigma^2} \right]$$

10.4.2 Posterior of δ

Proposition 10.4.2.

$$\delta \mid \mu, \sigma^2, y \sim \mathcal{N}(\delta_n, \tau_n^2)$$

$$\tau_n^2 = \left(\frac{1}{\tau_0^2} + \frac{n_1 + n_2}{\sigma^2} \right)^{-1}$$

$$\delta_n = \tau_n^2 \left[\frac{\delta_0}{\tau_0^2} + \frac{\sum (y_{i,1} - \mu)}{\sigma^2} - \frac{\sum (y_{i,2} - \mu)}{\sigma^2} \right]$$

10.4.3 Posterior of σ^2

Proposition 10.4.3.

$$\sigma^2 \mid \mu, \delta, y \sim \textit{Inv-Gamma} \left(\frac{\nu_n}{2}, \frac{\nu_n \sigma_n^2}{2} \right)$$

$$\nu_n = \nu_0 + n_1 + n_2$$

$$\nu_n \sigma_n^2 = \nu_0 \sigma_0^2 + \sum (y_{i,1} - (\mu + \delta))^2 + \sum (y_{i,2} - (\mu - \delta))^2$$

10.5 Gibbs Sampling Algorithm

Posterior inference about the scientifically meaningful contrast should use the transformed draws $2\delta^{(t)}$. For example, the posterior probability that group 1 exceeds group 2 by more than a clinically relevant amount c is estimated by

$$\frac{1}{T} \sum_{t=1}^{T} \mathbb{1}\{2\delta^{(t)} > c\}.$$

1. Initialize $\mu^{(0)}, \delta^{(0)}, \sigma^{2(0)}$
2. For $t = 1, \ldots, T$:
   (a) Sample:
   $$\mu^{(t)} \sim p(\mu \mid \delta^{(t-1)}, \sigma^{2(t-1)}, y)$$
   (b) Sample:
   $$\delta^{(t)} \sim p(\delta \mid \mu^{(t)}, \sigma^{2(t-1)}, y)$$
   (c) Sample:
   $$\sigma^{2(t)} \sim p(\sigma^2 \mid \mu^{(t)}, \delta^{(t)}, y)$$

This direct probability statement is different from testing the point null $\delta = 0$ and makes the decision threshold explicit.

10.6 Exchangeability

Definition 10.6.1. A sequence $Y_1, \ldots, Y_n$ is exchangeable if:

$$p(y_1, \ldots, y_n) = p(y_{\pi(1)}, \ldots, y_{\pi(n)})$$

for any permutation π.

Remark 10.6.1. Exchangeability is weaker than independence: it says labels carry no information about the joint law. De Finetti's theorem states that an infinite exchangeable Bernoulli sequence is a mixture of IID Bernoulli sequences,

$$p(y_1, \ldots, y_n) = \int_0^1 \prod_{i=1}^n \theta^{y_i}(1-\theta)^{1-y_i} \, \mathrm{d}G(\theta).$$

Thus conditional IID sampling is a representation of exchangeable dependence, not a claim that observations are marginally independent.

10.7 Hierarchical Normal Model

10.7.1 Model Structure

Definition 10.7.1. Within-group model:

$$Y_{ij} \mid \theta_j, \sigma^2 \sim \mathcal{N}(\theta_j, \sigma^2)$$

Between-group model:

$$\theta_j \mid \mu, \tau^2 \sim \mathcal{N}(\mu, \tau^2)$$

Remark 10.7.1.

- θ_j = group-specific mean
- μ = global mean
- τ^2 = between-group variance

10.8 Interpretation

Conditional on (μ, τ^2, σ^2) and with n_j observations in group j, conjugacy gives

$$\theta_j \mid y, \mu, \tau^2, \sigma^2 \sim \mathcal{N}(m_j, v_j), \quad v_j = \left(\frac{n_j}{\sigma^2} + \frac{1}{\tau^2}\right)^{-1},$$

where

$$m_j = w_j \bar{y}_j + (1 - w_j)\mu, \qquad w_j = \frac{n_j \tau^2}{\sigma^2 + n_j \tau^2}.$$

The group estimate is therefore a precision-weighted average of its sample mean and the population mean.

10.9 Shrinkage Effect

The weight w_j increases with group size and between-group variation and decreases with observation noise. If $\tau^2 \downarrow 0$, all group effects collapse toward the common mean. If $\tau^2 \to \infty$ or $n_j \to \infty$, $w_j \to 1$ and the group sample mean dominates.

Example 10.9.1. Suppose $\mu = 10$, $\sigma = 4$, $\tau = 2$, and a group of size $n_j = 4$ has $\bar{y}_j = 16$. Then $w_j = 4(4)/(16 + 4(4)) = 1/2$, so the posterior mean is $m_j = 13$. A group of size 36 with the same sample mean has $w_j = 144/(160) = 0.9$ and posterior mean 15.4. The smaller group is shrunk more because its mean is less precisely estimated.

10.10 Why Hierarchical Models Matter

Partial pooling can lower repeated-sampling mean squared error even though it introduces bias for an individual group. The gain comes from reducing the large variance of noisy group means. Posterior uncertainty also reflects both within-group noise and uncertainty in the population distribution. The hyperparameters must themselves be assigned priors; with few groups, the posterior for τ can be highly prior-sensitive.

10.11 Summary

Hierarchical modeling separates within-group noise from between-group heterogeneity and uses the learned population distribution to stabilize group estimates. The resulting shrinkage is adaptive: uncertain small groups borrow more information, while precise large groups retain more of their individual signal.

Chapter 11

Bayesian Linear Models

11.1 Model Setup

Definition 11.1.1. The Bayesian linear model is:

$$y = X\beta + \epsilon$$

$$\epsilon \sim \mathcal{N}(0, \sigma^2 I_n)$$

Remark 11.1.1.

- $y \in \mathbb{R}^n$
- $X \in \mathbb{R}^{n \times p}$
- $\beta \in \mathbb{R}^p$
- $\sigma^2 =$ noise variance

11.2 Likelihood

Proposition 11.2.1.

$$p(y \mid \beta, \sigma^2) \propto (\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2}(y - X\beta)^\top (y - X\beta)\right)$$

11.3 Conjugate Prior

Definition 11.3.1. Normal–Inverse-Gamma prior:

$$\beta \mid \sigma^2 \sim \mathcal{N}(\beta_0, \sigma^2 V_0)$$

$$\sigma^2 \sim \text{Inv-Gamma}\left(\frac{\nu_0}{2}, \frac{\nu_0 \sigma_0^2}{2}\right)$$

Remark 11.3.1.

- Prior scales with σ^2
- Enables closed-form posterior

11.4 Posterior Distribution

Completing the square gives

$$\begin{aligned}&(y - X\beta)^\top (y - X\beta) + (\beta - \beta_0)^\top V_0^{-1} (\beta - \beta_0) \\ &\quad = (\beta - \beta_n)^\top V_n^{-1} (\beta - \beta_n) + S_n,\end{aligned}$$

where

$$V_n^{-1} = V_0^{-1} + X^\top X, \quad \beta_n = V_n(V_0^{-1}\beta_0 + X^\top y),$$

and

$$S_n = y^\top y + \beta_0^\top V_0^{-1} \beta_0 - \beta_n^\top V_n^{-1} \beta_n.$$

This identity supplies the Gaussian conditional posterior for β and the updated inverse-gamma scale for σ^2.

11.4.1 Posterior of β

Proposition 11.4.1.

$$\beta \mid \sigma^2, y \sim \mathcal{N}(\beta_n, \sigma^2 V_n)$$

$$V_n = (V_0^{-1} + X^\top X)^{-1}$$

$$\beta_n = V_n(V_0^{-1}\beta_0 + X^\top y)$$

11.4.2 Posterior of σ^2

Proposition 11.4.2.

$$\sigma^2 \mid y \sim \textit{Inv-Gamma}\left(\frac{\nu_n}{2}, \frac{\nu_n \sigma_n^2}{2}\right)$$

$$\nu_n = \nu_0 + n$$

$$\nu_n \sigma_n^2 = \nu_0 \sigma_0^2 + (y - X\beta_n)^\top (y - X\beta_n) + (\beta_n - \beta_0)^\top V_0^{-1} (\beta_n - \beta_0)$$

11.5 Marginal Posterior of β

Proposition 11.5.1.

$$\beta \mid y \sim t_{\nu_n}(\beta_n, \sigma_n^2 V_n)$$

Remark 11.5.1.

- arises from integrating out σ^2
- heavier tails than normal

11.6 Posterior Predictive Distribution

Proposition 11.6.1. *For a new observation y^* with covariate x^*:*

$$y^* \mid y \sim t_{\nu_n}\left(x^{*\top}\beta_n, \sigma_n^2(1 + x^{*\top}V_n x^*)\right)$$

11.7 Interpretation

The posterior mean

$$\beta_n = (V_0^{-1} + X^\top X)^{-1}(V_0^{-1}\beta_0 + X^\top y)$$

combines prior and data in precision space. In more than one dimension it is not generally a scalar weighted average: the eigendirections of V_0^{-1} and $X^\top X$ may differ, so shrinkage depends on direction. Poorly identified directions of the design are influenced most strongly by the prior.

11.8 Connection to OLS

Proposition 11.8.1. *If prior is diffuse:*

$$V_0^{-1} \to 0$$

then:

$$\beta_n \to (X^\top X)^{-1} X^\top y$$

Remark 11.8.1. This limit requires $X^\top X$ to be invertible. When $p > n$ or the design is rank deficient, OLS is nonunique, while a proper positive-definite prior still produces a proper Gaussian conditional posterior.

11.9 Shrinkage Interpretation

To see shrinkage explicitly, suppose the columns of X are orthogonal, with

$$X^\top X = \mathrm{diag}(d_1, \ldots, d_p), \qquad V_0 = \lambda^{-1} I, \qquad \beta_0 = 0.$$

Then

$$(\beta_n)_j = \frac{d_j}{d_j + \lambda}(\hat{\beta}_{\mathrm{OLS}})_j.$$

Directions supported by little design information have small d_j and are shrunk most strongly.

11.10 Special Case: Ridge Regression

Proposition 11.10.1. *If $\beta \mid \sigma^2 \sim \mathcal{N}(0, \sigma^2\lambda^{-1}I)$, then the posterior mean and mode are*

$$\beta_n = (X^\top X + \lambda I)^{-1} X^\top y$$

and coincide with the ridge estimator.

Proof. Up to constants, the negative log posterior conditional on σ^2 is

$$\frac{1}{2\sigma^2}\left\{\|y - X\beta\|_2^2 + \lambda \|\beta\|_2^2\right\}.$$

Setting its gradient to zero gives $(X^\top X + \lambda I)\beta = X^\top y$. The scaling by σ^2 in the prior is essential for the penalty parameter to be exactly λ. □

11.11 Gibbs Sampling

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1. Initialize $\beta^{(0)}, \sigma^{2(0)}$
2. For $t = 1, \ldots, T$:
   (a) Sample:
   $$\beta^{(t)} \sim p(\beta \mid \sigma^{2(t-1)}, y)$$
   (b) Sample:
   $$\sigma^{2(t)} \sim p(\sigma^2 \mid \beta^{(t)}, y)$$

11.12 Summary

The normal–inverse-gamma prior makes the linear model analytically tractable: conditioning on σ^2 gives a Gaussian posterior, and integrating σ^2 gives Student distributions for coefficients and predictions. Diffuse-prior limits recover OLS when it is identifiable, while an isotropic Gaussian prior yields ridge regularization and remains well defined for a rank-deficient design.

Chapter 12

Generalized Linear Models (GLMs)

12.1 Motivation

Gaussian linear regression permits unbounded responses and constant conditional variance. Binary outcomes and counts violate both features. A generalized linear model specifies an exponential-family sampling law and uses a link function to map its constrained mean to an unconstrained linear predictor.

12.2 GLM Structure

A GLM has conditional density

$$p(y_i \mid \theta_i, \phi) = \exp\left\{\frac{y_i\theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi)\right\},$$

linear predictor $\eta_i = x_i^\top \beta$, and link $g(\mu_i) = \eta_i$, where $\mu_i = \mathbb{E}(Y_i \mid x_i) = b'(\theta_i)$. The variance is $\mathrm{Var}(Y_i \mid x_i) = a(\phi)b''(\theta_i)$, so the sampling family determines a mean–variance relationship rather than assuming constant variance.

12.3 Exponential Family Form

Definition 12.3.1.

$$p(y \mid \theta) = h(y)\exp\{\phi(\theta)t(y) - A(\theta)\}$$

Remark 12.3.1. GLMs arise because many common distributions (Bernoulli, Poisson, Normal) belong to this family

12.4 Canonical Link

Definition 12.4.1. A canonical link satisfies:

$$\eta = \phi(\theta)$$

Remark 12.4.1. With the canonical link, $\theta_i = x_i^\top \beta$. The score and observed curvature then take especially simple forms. Canonical links are convenient, but scientific interpretability or robustness may justify a different link.

12.5 Common GLMs

12.5.1 Logistic Regression (Binary Data)

Definition 12.5.1.

$$y_i \sim \text{Bernoulli}(p_i)$$

$$\log\left(\frac{p_i}{1-p_i}\right) = x_i^\top \beta$$

Proposition 12.5.1. *Likelihood:*

$$p(y \mid \beta) = \prod_i p_i^{y_i}(1-p_i)^{1-y_i}$$

The log likelihood is

$$\ell(\beta) = \sum_i \{y_i x_i^\top \beta - log(1 + e^{x_i^\top \beta})\}.$$

Its gradient is $X^\top(y-p)$ and its negative Hessian is $X^\top W X$, where $W = \text{diag}\{p_i(1-p_i)\}$. These expressions drive both Newton optimization and the Gaussian Laplace approximation.

Example 12.5.1 (Odds-ratio interpretation)**.** If a treatment indicator has coefficient $\beta_1 = \log 2$, then holding other covariates fixed doubles the odds of success. It does not generally double the success probability: moving from probability 0.1 gives $2(0.1)/(0.9+2(0.1)) \approx 0.182$, whereas moving from 0.5 gives 2/3.

12.5.2 Poisson Regression (Count Data)

Definition 12.5.2.

$$y_i \sim \text{Poisson}(\lambda_i)$$

$$\log(\lambda_i) = x_i^\top \beta$$

An exposure e_i is incorporated through an offset, $\log \lambda_i = \log e_i + x_i^\top \beta$. Then e^{β_j} is a rate ratio, not merely a ratio of raw counts.

Example 12.5.2 (Rate with unequal exposure)**.** Suppose two clinics record 8 and 12 events over 100 and 50 person-years. Modeling raw counts ignores that the second clinic has half the exposure. Using offsets compares rates 0.08 and 0.24; a clinic indicator coefficient near $\log 3$ represents the threefold rate ratio.

12.5.3 Gaussian GLM

Definition 12.5.3.

$$y_i \sim \mathcal{N}(x_i^\top \beta, \sigma^2)$$

Remark 12.5.1. This reduces to the standard linear model.

12.6 Bayesian GLMs

12.6.1 Prior

Definition 12.6.1.

$$\beta \sim \mathcal{N}(\beta_0, \Sigma_0)$$

12.7 Posterior

Proposition 12.7.1.

$$p(\beta \mid y) \propto p(y \mid \beta) p(\beta)$$

The Gaussian prior regularizes coefficients and makes the posterior proper in many cases where the likelihood alone is problematic. In completely separated logistic data, the MLE diverges because likelihood increases along a separating direction. A proper Gaussian prior supplies quadratic decay and produces a finite posterior mode and normalizing constant.

12.8 Inference Methods

12.8.1 Laplace Approximation

Definition 12.8.1. Approximate posterior by Gaussian around mode:

$$p(\beta \mid y) \approx \mathcal{N}(\hat{\beta}, H^{-1})$$

For logistic regression with prior covariance Σ_0, the negative Hessian at the mode is

$$H = X^\top \widehat{W} X + \Sigma_0^{-1}.$$

Thus the approximation automatically combines likelihood and prior precision. It is useful for a well-identified unimodal posterior but can understate tail uncertainty in small or separated samples.

12.8.2 Metropolis-Hastings

---

1. Propose $\beta^* \sim q(\cdot \mid \beta^{(t)})$
2. Accept with probability:

$$\min\left(1, \frac{p(y \mid \beta^*)p(\beta^*)}{p(y \mid \beta^{(t)})p(\beta^{(t)})}\right)$$

Remark 12.8.1. Used when posterior cannot be sampled directly

12.8.3 Gibbs Sampling (Augmentation)

For probit regression, latent Gaussian variables produce truncated-normal and Gaussian full conditionals, as derived in the next chapter. For logistic regression, Pólya–Gamma augmentation uses the identity

$$\frac{e^{\kappa\psi}}{(1+e^{\psi})^b} = 2^{-b}e^{\kappa\psi}\int_0^\infty e^{-\omega\psi^2/2}p_{PG}(\omega \mid b, 0)\,\mathrm{d}\omega,$$

where $\kappa = a - b/2$. Conditional on the latent ω_i, the likelihood is quadratic in β, giving a Gaussian update. The augmentation is exact; its purpose is computational, not an approximation to the logistic link.

12.9 Interpretation

Coefficients act linearly on the transformed mean. On the response scale, marginal effects depend on the covariate values: for logistic regression, $\partial p_i/\partial x_{ij} = p_i(1-p_i)\beta_j$. Posterior inference can therefore report odds ratios, predictive probabilities at meaningful covariate profiles, and average marginal effects; these answer different scientific questions.

12.9.1 Logistic Regression

Proposition 12.9.1.

$$\beta_j = \textit{log-odds change per unit increase in } x_j$$

12.9.2 Poisson Regression

Proposition 12.9.2.

$$e^{\beta_j} = \textit{multiplicative effect on expected count}$$

12.10 Summary

GLMs couple an exponential-family mean–variance relationship with a linked linear predictor. Bayesian priors regularize weak or separated designs, but usually destroy elementary conjugacy. Laplace approximation, Metropolis or Hamiltonian methods, and exact augmentation are alternative computational routes whose adequacy must be judged using posterior geometry and diagnostics.

Chapter 13

Bayesian Probit Regression Analysis

13.1 Model Setup

Definition 13.1.1. Binary response model:

$$y_i \in \{0, 1\}$$

$$\mathbb{P}(y_i = 1 \mid x_i) = \Phi(x_i^\top \beta)$$

Here Φ is the standard normal CDF. The link maps every real linear predictor to a probability in $(0, 1)$ and is especially convenient because it has a latent Gaussian representation.

13.2 Latent Variable Representation

Definition 13.2.1. Introduce latent variable:

$$z_i = x_i^\top \beta + \epsilon_i$$

$$\epsilon_i \sim \mathcal{N}(0, 1)$$

$$y_i = \begin{cases} 1 & \text{if } z_i > 0 \\ 0 & \text{if } z_i \leq 0 \end{cases}$$

Proposition 13.2.1.

$$\mathbb{P}(y_i = 1 \mid x_i) = \mathbb{P}(z_i > 0) = \Phi(x_i^\top \beta)$$

Remark 13.2.1. Probit regression is a thresholded Gaussian linear model.

13.3 Likelihood

Proposition 13.3.1.

$$p(y \mid \beta) = \prod_{i=1}^{n} \Phi(x_i^\top \beta)^{y_i} \left[1 - \Phi(x_i^\top \beta)\right]^{1-y_i}$$

Remark 13.3.1. No conjugate prior exists in this form.

13.4 Prior

Definition 13.4.1.

$$\beta \sim \mathcal{N}(\beta_0, \Sigma_0)$$

13.5 Posterior

Proposition 13.5.1.

$$p(\beta \mid y) \propto p(y \mid \beta) p(\beta)$$

Remark 13.5.1. The posterior has no closed form and therefore requires approximation or sampling.

13.6 Data Augmentation

Definition 13.6.1. Introduce latent variables:

$$z_i \mid \beta \sim \mathcal{N}(x_i^\top \beta, 1)$$

Proposition 13.6.1. *Conditional distributions:*

$$z_i \mid y_i, \beta \sim \begin{cases} \mathcal{N}(x_i^\top \beta, 1) \text{ truncated at } (0, \infty) & y_i = 1 \\ \mathcal{N}(x_i^\top \beta, 1) \text{ truncated at } (-\infty, 0] & y_i = 0 \end{cases}$$

Remark 13.6.1. This converts the model into a Gaussian regression problem.

13.7 Conditional Posterior of β

Proposition 13.7.1.

$$\beta \mid z \sim \mathcal{N}(\beta_n, V_n)$$

$$V_n = (X^\top X + \Sigma_0^{-1})^{-1}$$

$$\beta_n = V_n(X^\top z + \Sigma_0^{-1}\beta_0)$$

Remark 13.7.1. Conditional on z, this is identical to Bayesian linear regression.

13.8 Gibbs Sampling Algorithm

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1. Initialize $\beta^{(0)}$
2. For $t = 1, \ldots, T$:
   (a) Sample latent variables:
   $$z_i^{(t)} \sim p(z_i \mid y_i, \beta^{(t-1)})$$
   (b) Sample coefficients:
   $$\beta^{(t)} \sim p(\beta \mid z^{(t)})$$

Remark 13.8.1. Every update is an exact draw from a full conditional, so no Metropolis–Hastings correction is needed. This does not guarantee rapid mixing: strong posterior dependence between z and β can still produce autocorrelation.

Example 13.8.1 (One latent-variable update)**.** Suppose $x_i^\top \beta = -0.5$ but $y_i = 1$. Then $Z_i \mid y_i = 1, \beta$ is $Normal(-0.5, 1)$ truncated below at zero. Its mean is

$$-0.5 + \frac{\phi(0.5)}{1 - \Phi(0.5)} \approx 0.64,$$

which is positive as required by the observed class. The latent draw converts the binary observation into a continuously valued pseudo-response for the Gaussian regression update.

13.9 Posterior Prediction

Proposition 13.9.1.

$$\mathbb{P}(y^* = 1 \mid x^*, \beta) = \Phi(x^{*\top}\beta)$$

Remark 13.9.1. Posterior predictive:

$$\mathbb{P}(y^* = 1 \mid y) = \int \Phi(x^{*\top}\beta) p(\beta \mid y)\, d\beta$$

13.10 Interpretation

The coefficient β_j is an additive effect on the latent Gaussian index, not a log-odds effect. On the probability scale,

$$\frac{\partial}{\partial x_j}\mathbb{P}(Y = 1 \mid x, \beta) = \phi(x^\top \beta)\beta_j,$$

so the marginal effect depends on the full covariate profile.

13.11 Comparison with Logistic Regression

Logistic regression uses the logistic CDF, while probit uses the normal CDF. After rescaling coefficients, their fitted probabilities are often similar in the center but differ in the tails. Probit admits the truncated-normal Gibbs sampler above; logistic regression can use Metropolis, Hamiltonian Monte Carlo, or Pólya–Gamma augmentation. Computational convenience alone should not substitute for checking which link fits the application.

13.12 Summary

Probit regression represents a binary response as a thresholded Gaussian variable. Augmentation does not change the observed-data model; it exposes Gaussian conditional structure and enables Gibbs sampling. Coefficients act on a latent scale, so probability-scale effects should be reported at meaningful covariate values or averaged over the observed population.

Chapter 14

Missing Data and Imputation

14.1 Observed-Data Inference

Write the complete data as $Y = (Y_{\text{obs}}, Y_{\text{mis}})$ and let R record which components are observed. Bayesian inference targets the observed-data posterior

$$p(\theta \mid Y_{\text{obs}}, R) \propto p(\theta) \int p(Y_{\text{obs}}, Y_{\text{mis}}, R \mid \theta, \psi) p(\psi) \, \mathrm{d}Y_{\text{mis}} \, \mathrm{d}\psi,$$

where ψ parameterizes the missingness mechanism. Filling a blank with a single mean and analyzing the completed table as if it were observed omits the uncertainty represented by this integral. Complete-case analysis can also discard information and is generally unbiased only under additional conditions on the missingness process.

14.2 Missing Data Mechanisms

Let $p(R \mid Y, \psi)$ denote the response model. Missingness is *completely at random* (MCAR) when this distribution does not depend on Y. It is *missing at random* (MAR) when

$$p(R \mid Y_{\text{obs}}, Y_{\text{mis}}, \psi) = p(R \mid Y_{\text{obs}}, \psi).$$

It is *not missing at random* (MNAR) when the probability of observation continues to depend on an unobserved value after conditioning on observed variables.

MAR alone is not quite the statement that the mechanism can be ignored. One also needs *parameter distinctness*: the prior must factor as $p(\theta, \psi) = p(\theta)p(\psi)$. Under MAR and parameter distinctness, the factor involving R is constant in θ after conditioning on the observed data, so likelihood-based inference for θ may omit the response model. Under MNAR, the observed data usually do not identify the missingness mechanism without unverifiable assumptions; sensitivity analysis is therefore part of the inferential problem.

14.3 Data Augmentation

Augmenting the state with Y_{mis} gives the joint posterior

$$p(\theta, Y_{\text{mis}} \mid Y_{\text{obs}}) \propto p(\theta) p(Y_{\text{obs}}, Y_{\text{mis}} \mid \theta).$$

A two-block Gibbs sampler alternates

$$\theta^{(t)} \sim p(\theta \mid Y_{\text{obs}}, Y_{\text{mis}}^{(t-1)}), \qquad Y_{\text{mis}}^{(t)} \sim p(Y_{\text{mis}} \mid Y_{\text{obs}}, \theta^{(t)}).$$

Marginally, the retained draws of θ target $p(\theta \mid Y_{\text{obs}})$. An imputed value is therefore a posterior draw, not a best guess that should be treated as known.

Example 14.3.1 (Correlated Gaussian imputation)**.** Suppose $(Y_1, Y_2)^\top \sim \mathcal{N}_2(\mu, \Sigma)$, with

$$\mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \qquad \Sigma = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}.$$

If $Y_1 = y_1$ is observed and Y_2 is missing, then

$$Y_2 \mid Y_1 = y_1, \mu, \Sigma \sim \mathcal{N}\Big(\mu_2 + \rho\frac{\sigma_2}{\sigma_1}(y_1 - \mu_1), \sigma_2^2(1 - \rho^2)\Big).$$

Mean imputation would ignore both the predictive adjustment from y_1 and the residual uncertainty. When μ and Σ are unknown, drawing them inside the Gibbs sampler propagates parameter uncertainty as well.

14.4 Multiple Imputation

Multiple imputation stores m approximately independent completed datasets, analyzes each using the same complete-data procedure, and combines the results. If $\hat{q}_j$ has estimated variance U_j in completed dataset j, define

$$\bar{q} = \frac{1}{m}\sum_{j=1}^{m} \hat{q}_j, \quad \bar{U} = \frac{1}{m}\sum_{j=1}^{m} U_j, \quad B = \frac{1}{m-1}\sum_{j=1}^{m} (\hat{q}_j - \bar{q})^2.$$

Rubin's total variance estimate is $T = \bar{U} + (1 + 1/m)B$. The within-imputation term $\bar{U}$ measures ordinary sampling uncertainty; B measures uncertainty about the missing values. This decomposition explains why analyzing a single completed dataset yields intervals that are too narrow.

Chapter 15

GLM and Beyond

15.1 Limitations of GLMs

For a generalized linear model, the conditional mean satisfies $g\{\mathbb{E}(Y_i \mid x_i)\} = x_i^\top \beta$, and the conditional distribution is chosen from an exponential family. This imposes assumptions on the link, mean–variance relation, linear predictor, and conditional independence. Failure of any one has a different consequence. A misspecified link distorts the mean, overdispersion makes model-based standard errors too small, omitted dependence overstates effective sample size, and an inadequate linear predictor leaves systematic structure in the residuals. Bayesian inference propagates uncertainty within the chosen model; it does not make these structural assumptions automatically correct.

15.2 Non-Conjugacy

Even with a Gaussian prior, logistic and Poisson regression generally lack a closed-form posterior. For example, under $Y_i \sim \text{Bernoulli}(\text{logit}^{-1}(x_i^\top \beta))$ and $\beta \sim \mathcal{N}(0, \Sigma_0)$,

$$\log p(\beta \mid y) = \sum_i \{y_i x_i^\top \beta - \log(1 + e^{x_i^\top \beta})\} - \frac{1}{2}\beta^\top \Sigma_0^{-1} \beta + C.$$

The log posterior is concave when Σ_0 is positive definite, but its normalizing integral has no elementary form. This motivates Laplace approximations, Metropolis methods, Hamiltonian Monte Carlo, or augmentation.

15.3 Laplace Approximation

Let $\ell(\beta) = \log p(y, \beta)$, let $\hat{\beta}$ maximize ℓ, and set $H = -\nabla^2 \ell(\hat{\beta})$. A second-order Taylor expansion gives

$$\ell(\beta) = \ell(\hat{\beta}) - \frac{1}{2}(\beta - \hat{\beta})^\top H (\beta - \hat{\beta}) + o(\left\|\beta - \hat{\beta}\right\|^2).$$

Exponentiating and normalizing the quadratic approximation yields $p(\beta \mid y) \approx \mathcal{N}(\hat{\beta}, H^{-1})$. The same calculation gives the marginal-likelihood approximation

$$p(y) \approx p(y, \hat{\beta})(2\pi)^{p/2}|H|^{-1/2}.$$

Laplace approximation is accurate when posterior mass is concentrated near a single interior mode. It can fail for skewed, heavy-tailed, multimodal, or boundary-concentrated posteriors.

15.4 Data Augmentation

Data augmentation introduces latent variables Z such that $p(\beta, Z \mid y)$ has simpler full conditionals than $p(\beta \mid y)$. Probit regression uses a thresholded Gaussian Z_i, while logistic regression can use Pólya–Gamma variables. Augmentation does not change the marginal posterior because $p(\beta \mid y) = \int p(\beta, Z \mid y)\, \mathrm{d}Z$, but it changes the geometry and autocorrelation of the Markov chain. Conditional conjugacy therefore improves algebraic convenience without guaranteeing rapid mixing.

15.5 Hierarchical GLMs

For grouped data, a random-intercept model may take

$$g\{\mathbb{E}(Y_{ij} \mid b_j)\} = x_{ij}^{\top}\beta + b_j, \qquad b_j \mid \tau^2 \sim \mathcal{N}(0, \tau^2).$$

The shared distribution partially pools noisy group effects toward zero. As $\tau^2 \downarrow 0$ the model approaches complete pooling; as τ^2 grows, the groups behave more independently. The posterior learns the pooling strength rather than fixing it in advance. Centered parameterizations $b_j = \tau z_j$ can mix poorly when data are weak; the noncentered form $z_j \sim \mathcal{N}(0, 1)$ often improves computation in that regime.

15.6 Overdispersion

The Poisson model enforces $\mathrm{Var}(Y_i \mid x_i) = \mathbb{E}(Y_i \mid x_i) = \lambda_i$. A gamma random effect produces a negative-binomial marginal distribution. If

$$Y \mid \Lambda \sim \mathrm{Poisson}(\Lambda), \qquad \Lambda \sim \mathrm{Gamma}(r, r/\mu)$$

in shape–rate form, then the laws of total expectation and variance give

$$\mathbb{E}(Y) = \mu, \qquad \mathrm{Var}(Y) = \mu + \frac{\mu^2}{r}.$$

The additional quadratic term captures unobserved heterogeneity. This derivation also gives the negative binomial a generative interpretation, rather than treating it merely as a variance correction.

15.7 Robust Models

Replacing Gaussian errors by a Student distribution limits the influence of large residuals. The scale-mixture representation

$$Y_i \mid \mu, \sigma^2, \lambda_i \sim \mathcal{N}(\mu, \sigma^2/\lambda_i), \qquad \lambda_i \sim \text{Gamma}(\nu/2, \nu/2)$$

implies $Y_i \mid \mu, \sigma^2 \sim t_\nu(\mu, \sigma)$. A large residual drives the posterior of λ_i downward and inflates its conditional variance, automatically reducing that observation's leverage. Heavy-tailed priors solve a different problem: they make large coefficient values less strongly penalized but do not themselves robustify the observation model.

15.8 Nonparametric Extensions

Bayesian nonparametric models place priors on infinite-dimensional objects. For example, a Dirichlet process mixture writes

$$Y_i \mid \theta_i \sim k(\, cdot \mid \theta_i), \qquad \theta_i \mid G \overset{\text{iid}}{\sim} G, \qquad G \sim \text{DP}(\alpha, G_0).$$

After integrating out G, an existing cluster with size n_k receives the next observation with probability proportional to n_k, while a new cluster is created with probability proportional to α. The effective number of occupied components grows with the data, so flexibility is not fixed by a finite parameter count.

15.9 Summary

Moving beyond a basic GLM requires identifying which assumption failed. Laplace approximation and augmentation address computation; random effects address dependence and heterogeneity; negative-binomial mixtures address overdispersion; Student likelihoods address outliers; and nonparametric priors address distributional rigidity. These modifications are not substitutes for one another, even though several can appear in the same hierarchical model.

Chapter 16

Model Selection

16.1 Motivation

Model comparison can target different questions. Posterior model probabilities compare a specified list of probabilistic models; predictive criteria ask which model predicts future observations well; posterior predictive checks ask whether a model reproduces scientifically relevant features at all. These goals should not be collapsed into a single ranking.

16.2 Bayesian Model Comparison

Assign prior probabilities $p(M_j)$ and within-model priors $p(\theta_j \mid M_j)$. Bayes' rule gives

$$p(M_i \mid y) = \frac{p(y \mid M_i)p(M_i)}{\sum_j p(y \mid M_j)p(M_j)}$$

for mutually exclusive candidate models. The posterior odds decompose as

$$\frac{p(M_1 \mid y)}{p(M_2 \mid y)} = \underbrace{\frac{p(y \mid M_1)}{p(y \mid M_2)}}_{BF_{12}} \frac{p(M_1)}{p(M_2)}.$$

Thus a Bayes factor updates model odds; it is not itself a posterior probability.

16.3 Marginal Likelihood

The marginal likelihood is the prior predictive density of the observed sample,

$$p(y \mid M) = \int p(y \mid \theta, M)p(\theta \mid M)\ \mathrm{d}\theta.$$

It averages likelihood over the prior rather than evaluating it at a fitted parameter. A diffuse prior spreads mass across values that predict the observed data poorly, reducing the average; this is the source of the so-called Occam factor and also the source of strong prior sensitivity.

Example 16.3.1 (Beta–binomial marginal likelihood)**.** Under $Y \mid \theta, M \sim \text{Binomial}(n, \theta)$ and $\theta \mid M \sim \text{Beta}(a, b)$,

$$p(y \mid M) = \binom{n}{y} \frac{B(a + y, b + n - y)}{B(a, b)}.$$

This follows by multiplying the binomial likelihood by the beta density and recognizing the beta integral. It is a complete evidence calculation, not a plug-in likelihood at the posterior mean.

16.4 Bayes Factors

For two models, $BF_{12} = p(y \mid M_1)/p(y \mid M_2)$. Values above one move prior odds toward M_1 and values below one move them toward M_2. Conventional labels such as "strong" evidence are context-dependent and hide the role of prior model odds, losses, and prior sensitivity. Reporting $\log BF_{12}$ together with the priors used is more informative than a label alone.

16.5 Important Considerations

An improper prior is specified only up to an arbitrary multiplicative constant. That constant cancels from a within-model posterior but not from a marginal likelihood, so ordinary Bayes factors are undefined. Even proper but increasingly diffuse priors can force a Bayes factor toward a point null while the classical test rejects it; this is the Jeffreys–Lindley phenomenon. Prior predictive simulation is therefore an essential model-comparison diagnostic: if the prior assigns negligible mass to scientifically plausible datasets, its Bayes factor is not a neutral measure of fit.

16.6 Alternative Criteria

16.6.1 BIC

For a regular k-dimensional model with fixed prior density positive near the MLE, Laplace's method gives

$$\log p(y \mid M) = \log p(y \mid \hat{\theta}, M) - \frac{k}{2} \log n + O_p(1).$$

Multiplying by -2 and dropping bounded model-dependent terms yields

$$\text{BIC} = -2 \log p(y \mid \hat{\theta}, M) + k \log n.$$

BIC is therefore a large-sample approximation under regular fixed-dimensional conditions, not a universal replacement for a marginal likelihood. Mixtures, singular latent-variable models, boundary parameters, and growing dimension can violate its derivation.

16.6.2 Deviance Information Criterion (DIC)

With deviance $D(\theta) = -2 \log p(y \mid \theta) + C$, one common definition is

$$p_D = \mathbb{E}\{D(\theta) \mid y\} - D\{\mathbb{E}(\theta \mid y)\}, \qquad \text{DIC} = D\{\mathbb{E}(\theta \mid y)\} + 2p_D.$$

DIC attempts to correct in-sample fit by an effective parameter count. It can depend on parameterization and behave poorly for multimodal or hierarchical posteriors, because the posterior mean may lie in a low-density region.

16.6.3 Log Pseudo-Marginal Likelihood (LPML)

Define the conditional predictive ordinate

$$\text{CPO}_i = p(y_i \mid y_{-i}) = \int p(y_i \mid \theta) p(\theta \mid y_{-i}) \, \mathrm{d}\theta.$$

Then $\text{LPML} = \sum_i \log \text{CPO}_i$. If $\theta^{(s)} \sim p(\theta \mid y)$, the harmonic identity

$$\text{CPO}_i^{-1} = \mathbb{E}\left[\frac{1}{p(y_i \mid \theta)} \middle| y \right]$$

permits estimation from full-posterior draws. Like other harmonic-mean identities it may be unstable when inverse likelihood terms have heavy tails; Pareto-smoothed importance sampling provides a more diagnostic modern approach to leave-one-out prediction.

16.7 Posterior Predictive Checks

The posterior predictive distribution is

$$p(y^{\text{rep}} \mid y) = \int p(y^{\text{rep}} \mid \theta) p(\theta \mid y) \, \mathrm{d}\theta.$$

Choose a discrepancy $T(y)$ tied to the scientific feature the model should reproduce, simulate replicated datasets, and compare $T(y)$ with $T(y^{\text{rep}})$. For Poisson counts, useful discrepancies include the fraction of zeros and the sample variance; a model can reproduce the mean while failing badly on both. Posterior predictive checks diagnose absolute misfit but do not supply posterior probabilities that a model is true.

16.8 Tradeoff: Fit vs Complexity

Marginal likelihood rewards prior predictive concentration around the entire observed dataset, whereas leave-one-out criteria reward predictive accuracy for held-out observations. They can rank models differently because they answer different questions. Model averaging can preserve uncertainty over the candidate list, but it cannot correct the omission of every plausible model. Selection should therefore be accompanied by predictive diagnostics and sensitivity analysis rather than treated as a terminal decision.

16.9 Summary

Bayes factors compare prior predictive distributions and require meaningful proper priors. BIC is a regular asymptotic approximation, while DIC and LOO criteria target predictive fit through different complexity corrections. Posterior predictive checks answer the separate question of whether the model can reproduce relevant features of the observed data. A defensible analysis states which of these questions is being asked and examines sensitivity to the modeling choices that determine the answer.

Chapter 17

Bayesian Hypothesis Testing and Multiplicity

17.1 Point Null Hypotheses

A point null $H_0 : \theta = \theta_0$ cannot receive positive posterior probability under a continuous prior alone. Bayesian testing therefore uses a mixture prior

$$\pi(\theta) = \omega_0 \delta_{\theta_0}(\theta) + (1 - \omega_0)\pi_1(\theta).$$

The posterior null probability is

$$\mathbb{P}(H_0 \mid y) = \frac{\omega_0 m_0(y)}{\omega_0 m_0(y) + (1 - \omega_0) m_1(y)},$$

where m_j is the marginal likelihood under H_j. This makes clear that a Bayesian point-null test requires both prior model odds and a prior under the alternative.

17.2 Normal-Mean Bayes Factor

Suppose $\bar{Y} \mid \theta \sim \mathcal{N}(\theta, \sigma^2/n)$ with known variance. Under H_0, $heta = 0$; under H_1, let $\theta \sim \mathcal{N}(0, \tau^2)$. Convolution gives

$$m_0(\bar{y}) = \phi(\bar{y}; 0, \sigma^2/n), \qquad m_1(\bar{y}) = \phi(\bar{y}; 0, \tau^2 + \sigma^2/n),$$

and hence

$$BF_{01} = \sqrt{1 + \frac{n\tau^2}{\sigma^2}} \exp\left[-\frac{z^2}{2} \frac{n\tau^2}{\sigma^2 + n\tau^2} \right], \qquad z = \frac{\sqrt{n}\bar{y}}{\sigma}.$$

For fixed z, increasingly diffuse τ favors the null through the leading Occam factor. This calculation is the simplest demonstration that a vaguely specified alternative is not neutral.

17.3 Testing by Decisions

Posterior probabilities do not themselves specify an action. If accepting H_1 when H_0 is true costs L_{10} and accepting H_0 when H_1 is true costs L_{01}, choose H_1 when

$$\mathbb{P}(H_0 \mid y)L_{10} < \mathbb{P}(H_1 \mid y)L_{01}.$$

Equivalently, the posterior odds threshold is the loss ratio. Scientific and regulatory contexts can therefore use the same posterior evidence but reach different actions because their consequences differ.

17.4 Multiple Testing

For hypotheses H_{0j}, let $q_j = \mathbb{P}(H_{0j} \mid y)$. If rejecting j incurs unit loss when its null is true and retaining it incurs loss λ when its alternative is true, the posterior expected loss separates across tests. Reject H_{0j} when

$$q_j < \frac{\lambda}{1+\lambda}.$$

Hierarchical priors share information about the prevalence and effect-size distribution across tests. This produces adaptive multiplicity correction: when the data suggest that signals are rare, posterior null probabilities increase globally without applying an external correction to each p-value.

Example 17.4.1 (Two-groups model)**.** Let latent indicators satisfy $\gamma_j \mid \omega \overset{\text{iid}}{\sim} \text{Bernoulli}(\omega)$, $\omega \sim \text{Beta}(a, b)$, and let observed estimates arise from either a null or signal distribution according to γ_j. The conditional update

$$\omega \mid \gamma \sim \text{Beta}\left(a + \sum_j \gamma_j, b + m - \sum_j \gamma_j\right)$$

shows how evidence from all m tests learns the signal prevalence and feeds back into every local posterior probability.

Chapter 18

Bayesian Variable Selection and Model Averaging

18.1 Spike-and-Slab Priors

In linear regression, introduce inclusion indicators γ_j and set

$$\beta_j \mid \gamma_j \sim (1-\gamma_j)\delta_0 + gamma_j\mathcal{N}(0,\tau^2), \qquad \gamma_j \mid \omega \overset{\text{iid}}{\sim} \text{Bernoulli}(\omega).$$

The posterior jointly represents uncertainty about model dimension and coefficient magnitude. Marginal inclusion probabilities $\mathbb{P}(\gamma_j = 1 \mid y)$ are often more stable than selecting the single highest-probability model, especially with correlated predictors.

18.2 The g-Prior

For a model with design X_γ, Zellner's prior is

$$\beta_\gamma \mid \sigma^2, g, M_\gamma \sim \mathcal{N}\{0, g\sigma^2(X_\gamma^\top X_\gamma)^{-1}\}.$$

It scales prior covariance to the likelihood geometry and is invariant to linear rescaling of predictors. Conditional on $sigma^2$, the posterior mean is

$$\mathbb{E}(\beta_\gamma \mid y, M_\gamma, g) = \frac{g}{1+g}\hat{\beta}_\gamma,$$

so g has a direct shrinkage interpretation. Fixed, empirical-Bayes, and mixture priors on g encode different complexity penalties.

18.3 Posterior Model Probabilities

For a finite model set,

$$p(M_\gamma \mid y) \propto p(y \mid M_\gamma)p(M_\gamma).$$

A uniform prior over all 2^p subsets does not imply a uniform prior over model size: it concentrates size near $p/2$. A beta–binomial hierarchy on the inclusion probability can represent sparsity and account for uncertainty about model size.

When p is moderate, a sampler can propose adding, deleting, or swapping one predictor and accept the move using posterior model odds. For large p, the model space is itself multimodal; local exploration diagnostics are as important here as in continuous-parameter MCMC.

18.4 Bayesian Model Averaging

Predictions average over structural uncertainty:

$$p(y^* \mid y) = \sum_{\gamma} p(y^* \mid y, M_\gamma) p(M_\gamma \mid y).$$

The total predictive variance decomposes into average within-model variance plus between-model variance of predictive means. Selecting one model omits the second term and can give intervals that are too narrow.

Example 18.4.1 (Correlated predictors)**.** Suppose two nearly collinear biomarkers are individually predictive but add little information together. The posterior may split mass between the two single-variable models, giving neither biomarker inclusion probability near one. Model-averaged predictions can nevertheless be stable because both models make similar predictions. This distinguishes uncertainty about an explanation from uncertainty about an outcome.

Chapter 19

Finite Mixture Models and Latent Allocations

19.1 Mixture Representation

A K-component mixture has density

$$p(y_i \mid \pi, \vartheta) = \sum_{k=1}^{K} \pi_k f(y_i \mid \vartheta_k), \qquad \sum_k \pi_k = 1.$$

Introduce $Z_i \mid \pi \sim \text{Categorical}(\pi)$ and $Y_i \mid Z_i = k \sim f(\cdot \mid \vartheta_k)$. Marginalizing Z_i recovers the mixture likelihood, while conditioning on allocations factorizes it by component.

19.2 Conjugate Gaussian Mixture Updates

With $\pi \sim \text{Dirichlet}(\alpha_1, \dots, \alpha_K)$, known common variance σ^2, and $\mu_k \sim \mathcal{N}(m_0, s_0^2)$, Gibbs updates are

$$\mathbb{P}(Z_i = k \mid -) \propto \pi_k \phi(y_i; \mu_k, \sigma^2),$$
$$\pi \mid Z \sim \text{Dirichlet}(\alpha_1 + n_1, \dots, \alpha_K + n_K),$$

and

$$\mu_k \mid Z, y \sim \mathcal{N}(m_k, s_k^2), quads_k^{-2} = s_0^{-2} + n_k \sigma^{-2}, quadm_k = s_k^2 \left(s_0^{-2} m_0 + sigma^{-2} \sum_{i:Z_i=k} y_i \right).$$

Empty components revert to their priors, which is essential for a valid sampler.

19.3 Label Switching

If priors are exchangeable across components, every permutation of component labels gives the same likelihood and prior. The posterior therefore has $K!$ symmetric modes. A chain

that switches labels is not failing; summaries such as the raw posterior mean of μ_1 become meaningless because the label has no scientific identity. One should summarize label-invariant quantities, impose a scientifically defensible identifying constraint, or post-process draws by a relabeling criterion.

19.4 Mixture Prediction

For a new observation,

$$p(y^* \mid y) = \int \sum_{k=1}^{K} \pi_k f(y^* \mid \vartheta_k) p(\pi, \vartheta \mid y) \, \mathrm{d}\pi \, \mathrm{d}\vartheta.$$

This integrates uncertainty in weights, component parameters, and allocation. Assigning a new observation to the nearest posterior-mean component ignores all three sources.

Example 19.4.1 (Ambiguous allocation)**.** In a two-component Gaussian mixture with equal weights, means -1 and 1, and unit variance, an observation at zero has equal posterior allocation probability for both components. A hard clustering reports arbitrary certainty, whereas the latent allocation posterior correctly records the ambiguity.

Chapter 20

Bayesian Workflow and Sensitivity Analysis

20.1 Prior Predictive Workflow

A prior is best assessed through the data it predicts. Simulate $\theta^{(s)} \sim p(\theta)$ and $y^{(s)} \sim p(y \mid \theta^{(s)})$, then inspect summaries on the same scale as the scientific measurements. Impossible counts, implausible ranges, or degenerate correlations reveal prior problems before the observed data can hide them.

20.2 Simulation-Based Calibration

Simulation-based calibration checks an inference algorithm under the assumed model. Repeatedly simulate $\theta^{(r)}$ and $y^{(r)}$ from the joint prior predictive law, fit the posterior, and record the rank of the generating parameter among posterior draws. Under exact computation and exchangeability, these ranks are discrete uniform. U-shaped histograms indicate underdispersed posteriors; hump-shaped histograms indicate overdispersion; skew can indicate bias. This validates implementation under the model, not whether the model is scientifically adequate.

20.3 Posterior Predictive Diagnostics

A useful discrepancy targets a model assumption. For count data, compare zero frequency and upper-tail counts; for regression, compare residual trends and groupwise variance; for time series, compare autocorrelations. A single omnibus discrepancy can average away a scientifically important failure.

Posterior predictive p-values

$$p_B = \mathbb{P}\{T(Y^{\text{rep}}, \theta) \geq T(Y, \theta) \mid y\}$$

are generally conservative because the observed data help construct both the posterior and the comparison. They are graphical diagnostic summaries, not uniform classical p-values under a point null.

20.4 Local Prior Sensitivity

Let a baseline prior be exponentially tilted, $\pi_\epsilon(\theta) \propto \pi_0(\theta) e^{\epsilon h(\theta)}$. For a posterior functional g, differentiation at zero gives

$$\frac{\mathrm{d}}{\mathrm{d}\epsilon} \mathbb{E}_\epsilon\{g(\theta) \mid y\}\Big|_{\epsilon=0} = \mathrm{Cov}_0\{g(\theta), h(\theta) \mid y\}.$$

Large posterior covariance identifies directions in which a small prior perturbation changes the answer. Global sensitivity complements this local calculation by refitting under substantively different plausible priors.

20.5 Communicating Robust Conclusions

A conclusion is robust when it persists across reasonable likelihoods, priors, computational settings, and predictive checks. Sensitivity need not be hidden: it identifies which assumptions are carrying information that the data do not contain. Reports should separate posterior uncertainty under one model, Monte Carlo error, and variation across modeling choices.